ECON 100B: Macroeconomics

Class 7: The Solow Model 2 (Jones Chapter 5)

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February 10, 2026

Agenda

- Announcements
- Current events

- Finish up Chapter 4: A Model of Production
- Chapter 5: The Solow Growth Model

Announcements (same from Class 05)

- Problem Set 1 is due on Gradescope on Friday 2/13

- Problem Set 2 will be due on Gradescope on Friday 2/27

- In-class Midterm the next Thursday 3/5

- Take-home Midterm Essay released 3/8, due 3/20

- Spring Break 3/23-3/27

Read both of these newspapers

- Wall Street Journal
- New York Times
- For access to both, see this Library guide

- You might skip the op-ed pages unless you like being horrified or vindicated, depending on your point of view

- But sometimes there are very good op-ed pieces in both

Current events

First half of ECON 100B

- Chapter 2 shows how we <u>measure</u> income, Y = C + I + G + NX

- **Chapters 3–6 discuss <u>long-run growth</u>**

  - Chapter 3 introduced the growth rate rules

$$g[X \times Y] = g[X] + g[Y] \qquad\qquad g[X/Y] = g[X] - g[Y]$$

$$g[X^a \times Y^b] = a \cdot g[X] + b \cdot g[Y]$$

  - Chapter 4 introduces Cobb-Douglas production

$$Y = A\ K^{1/3} L^{2/3}$$

with constant returns to scale
and declining marginal productivity of K or L

  - **Chapter 5 develops the Solow Model**

  - Chapter 6 develops the Romer Model

ALERT: We will use “old school” Romer in Chapter 6

- Jones 6e introduces a new Romer model with slightly different math and a slightly different solution. All are a little bit harder

- Students are welcome to read that version in 6e

- But PS 2 and exams and past exams will use “old school” Romer from Jones 1e-5e

- We will post a PDF of Chapter 6 from Jones 1e

Learning objectives in Chapter 5

- Investment in new capital comes from **saving**, which usually comes from domestic income not consumed: $S = Y - C = I$

- If people work and save at rate s out of income, then $S = sY$

- But capital also breaks at rate d

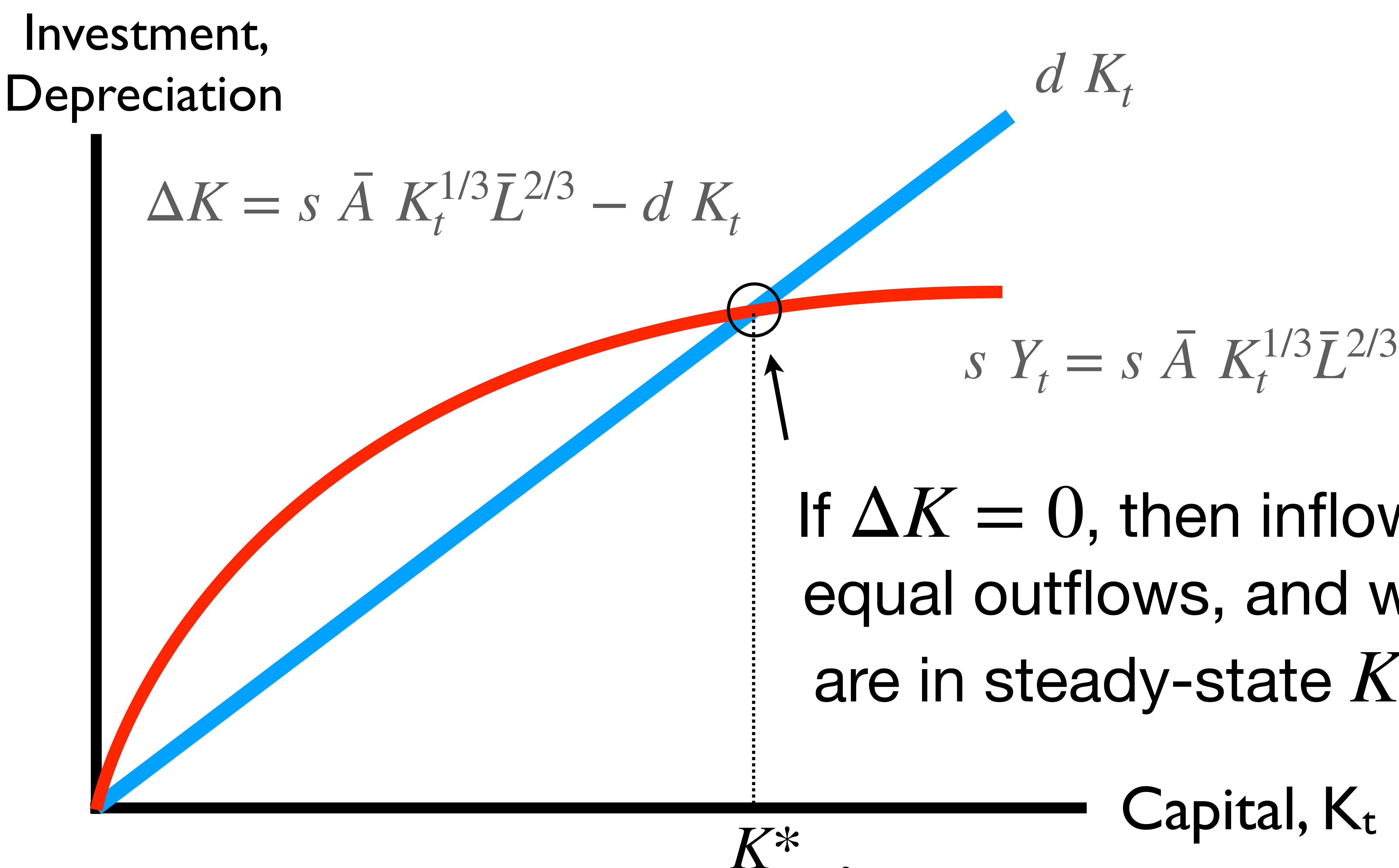
- The Solow Model includes 2 curves: **saving** sY, and **depreciation** dK,

  and where they cross, $sY = dK$, capital is unchanging at $K*$ in what we call a **steady state**

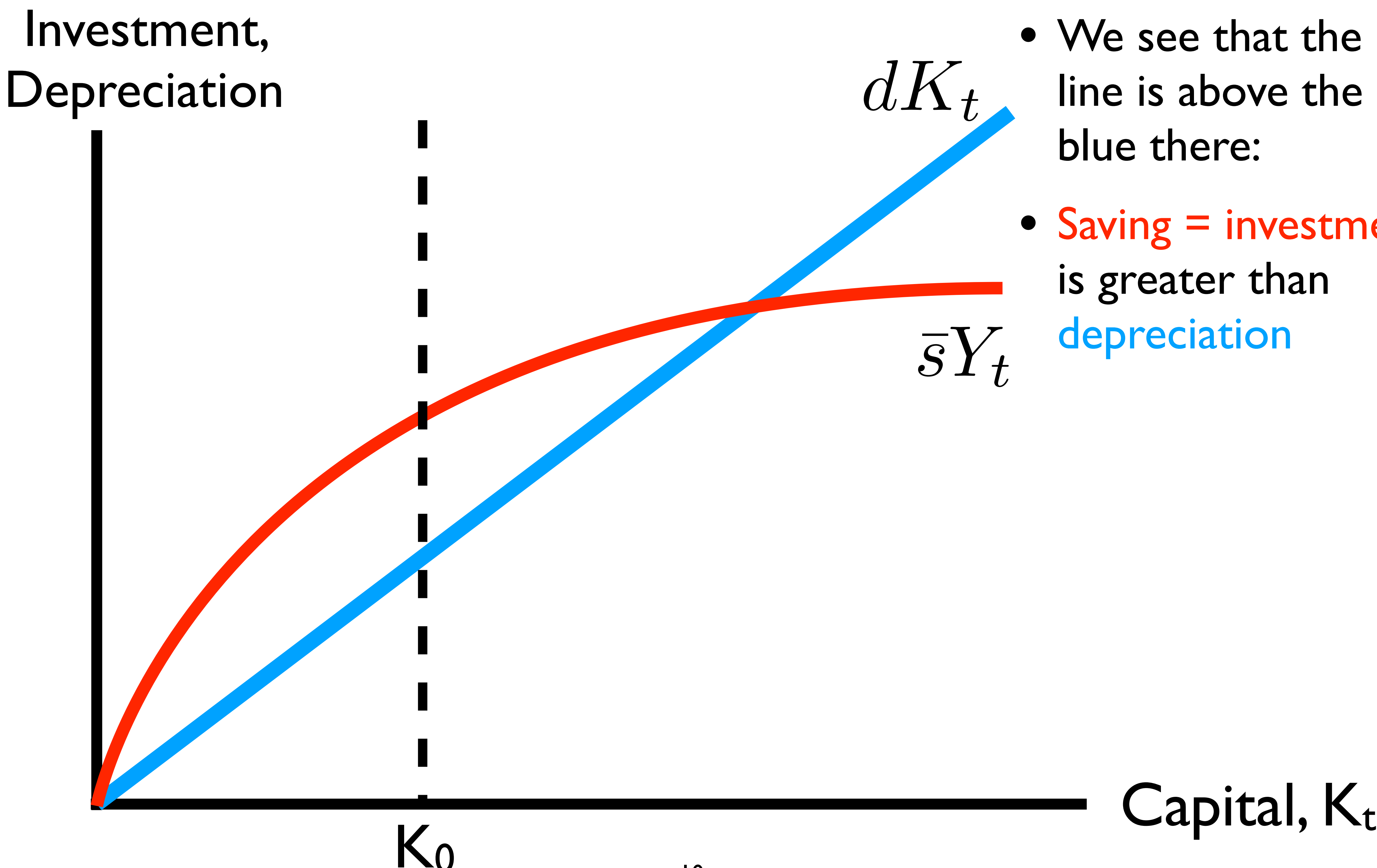
- When countries are far below their steady states, they **grow fast**

- A simple extension in §5.10 is a steady state with $k* = \dfrac{K}{L}$ with population growth at rate n

The curves cross at the **STEADY STATE**



Suppose the economy starts at this K_0:



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- Saving = investment is greater than depreciation

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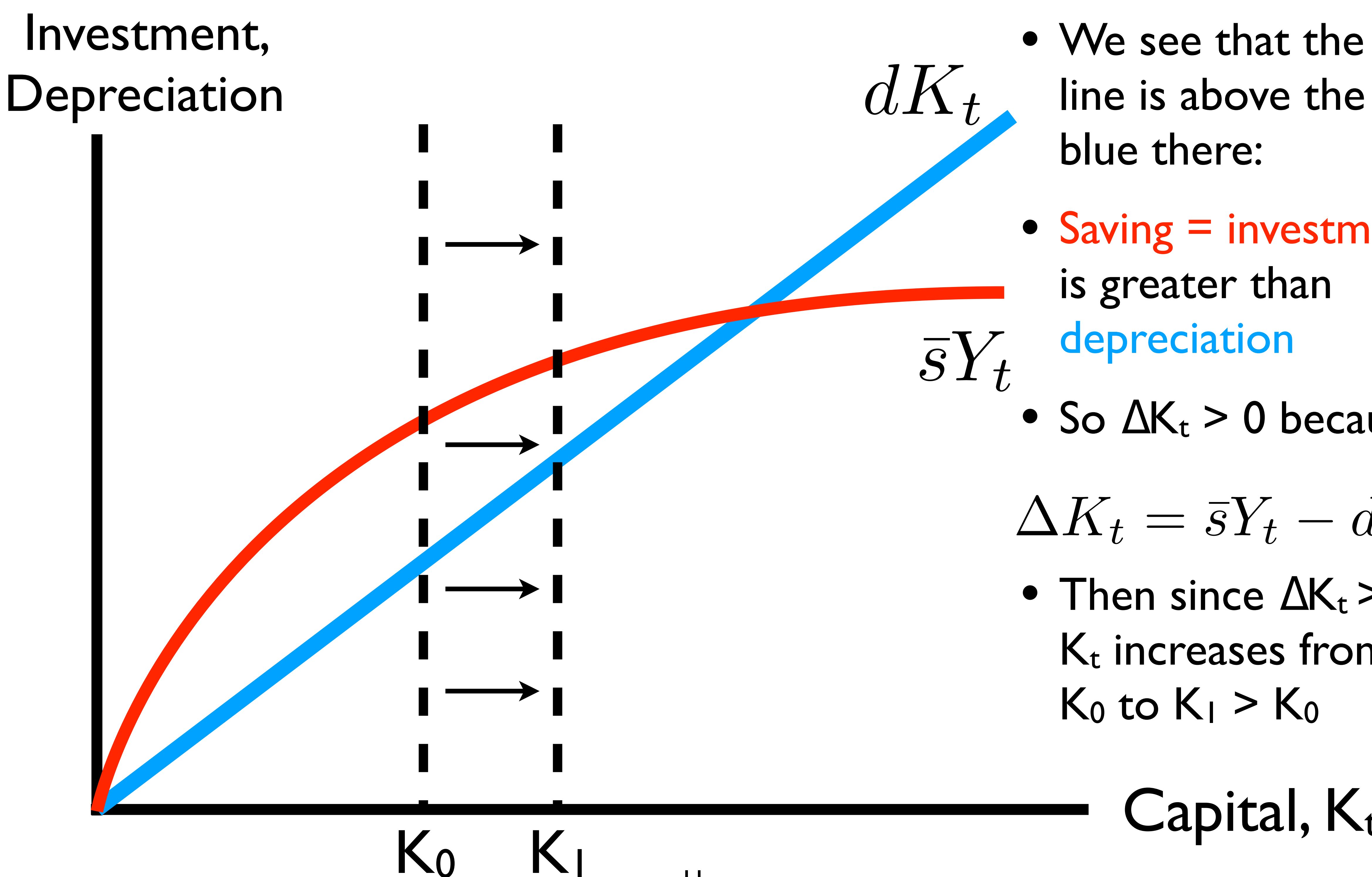
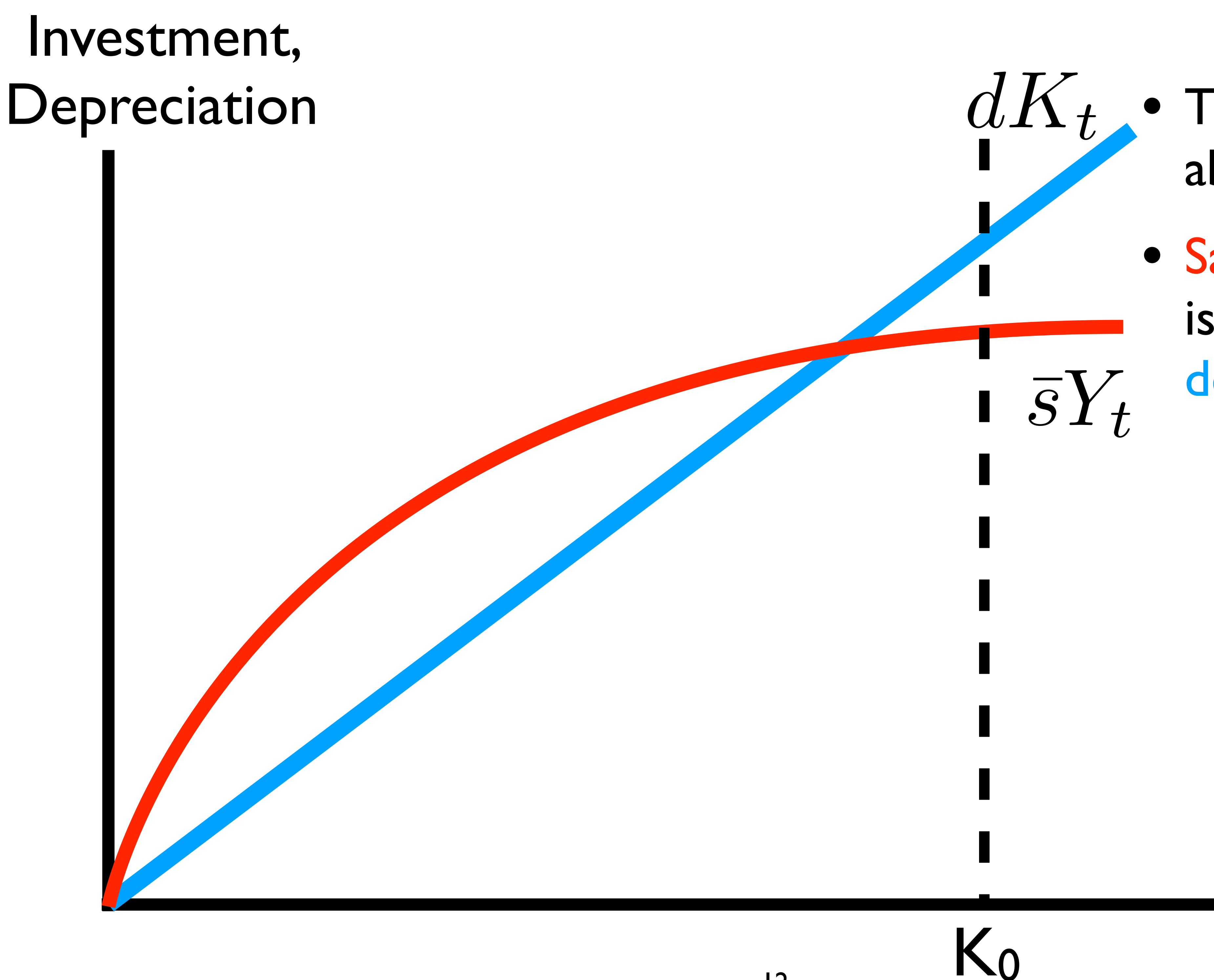


- We see that the red line is above the blue there:
- Saving = investment is greater than depreciation
- So $\Delta K_t > 0$ because

$$\Delta K_t = \bar{s}Y_t - dK_t$$

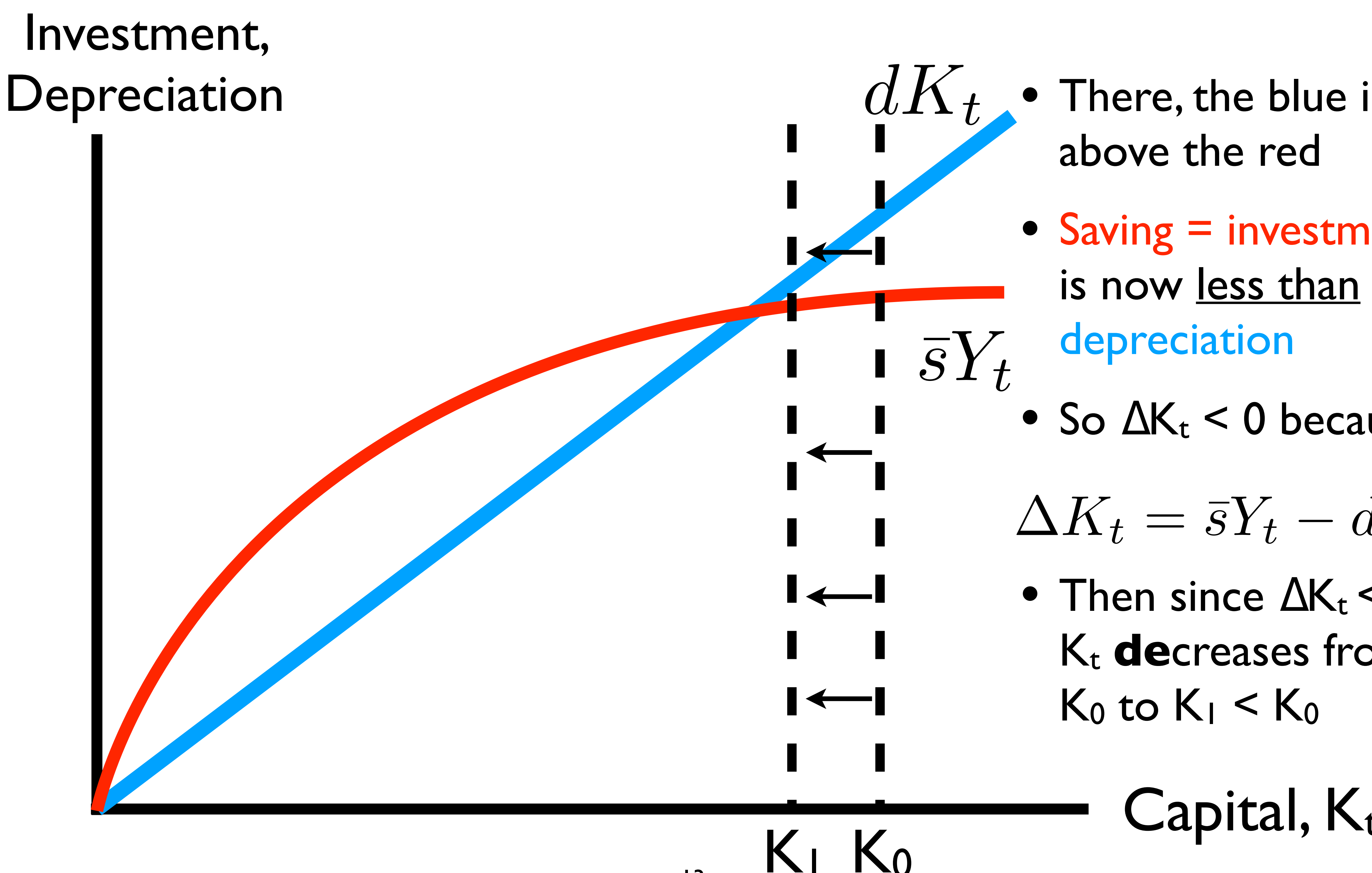
- Then since $\Delta K_t > 0$, K_t increases from K_0 to $K_1 > K_0$

Now imagine if we start at a K_0 **here**



- There, the blue is above the red
- Saving = investment is now <u>less than</u> depreciation

Now imagine if we start at a K_0 **here**



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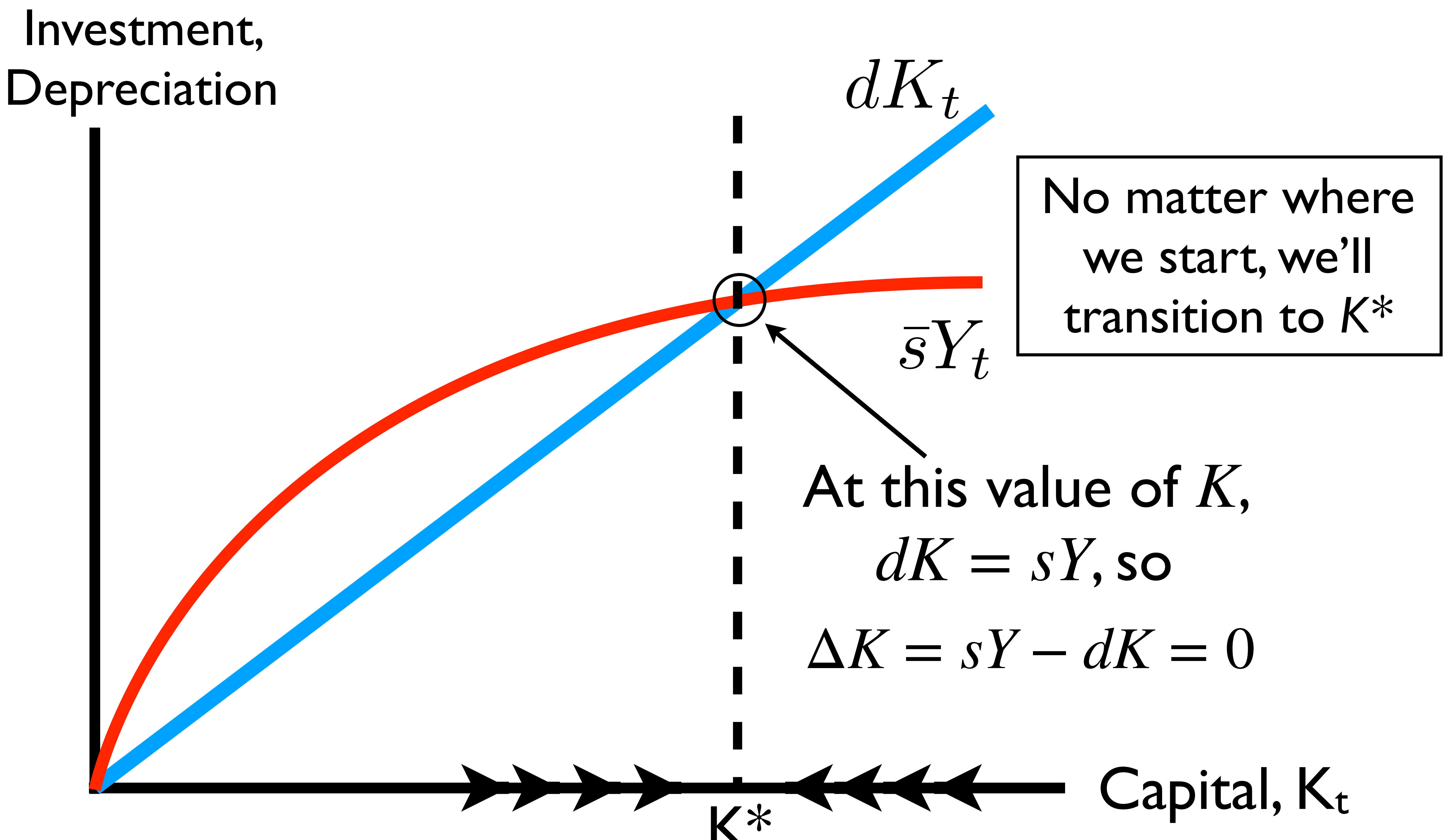
- Saving = investment is now less than depreciation
- So $\Delta K_t < 0$ because

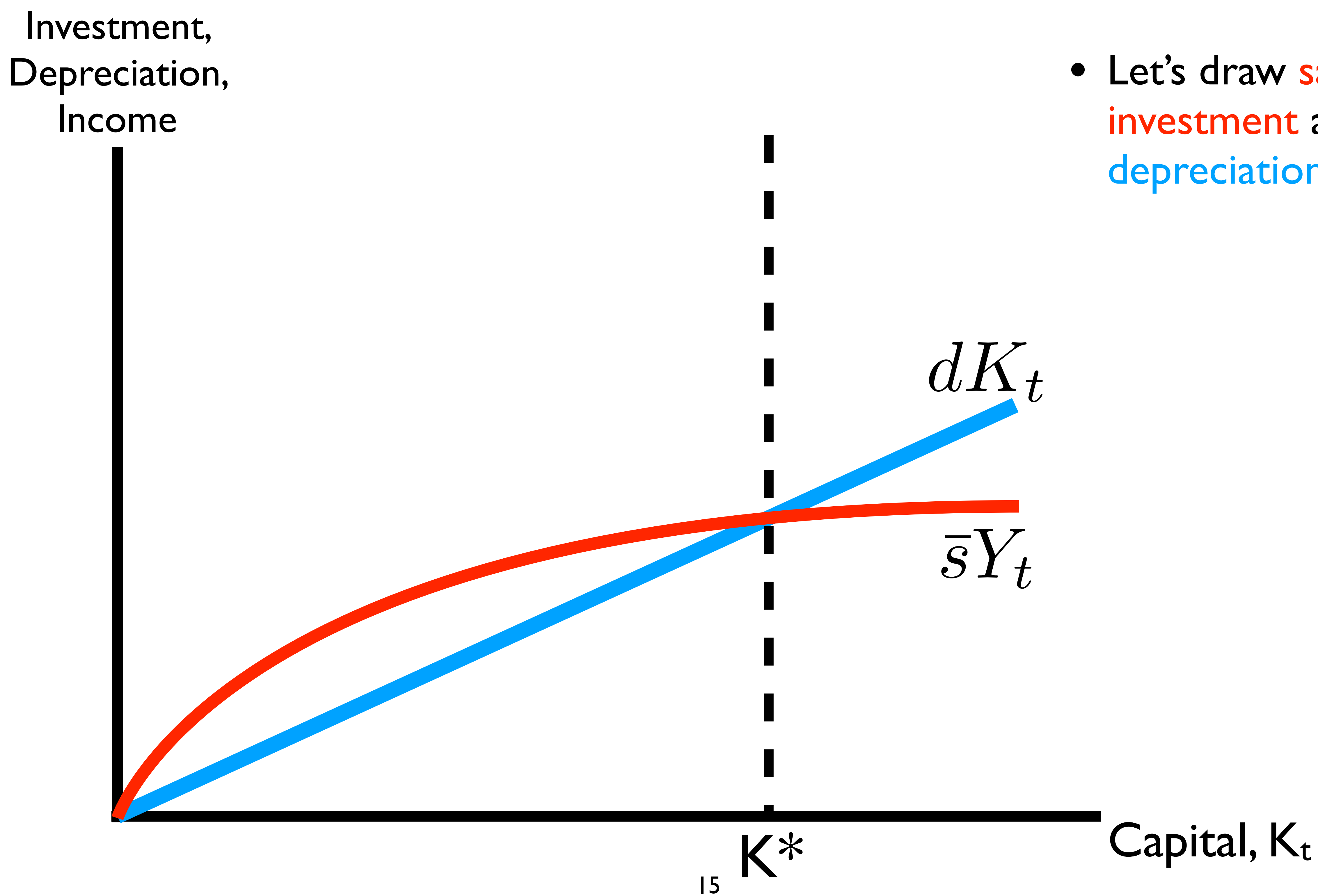
$$\Delta K_t = \bar{s}Y_t - dK_t$$

- Then since $\Delta K_t < 0$, K_t **de**creases from K_0 to $K_1 < K_0$

We call this the process of <u>transition dynamics</u>: Transitioning from any K_t toward the economy's *steady-state* K^*, where $\Delta K_t = 0$



We can also plot **output, Y,** which appears above everything



- Let's draw saving = investment and depreciation here

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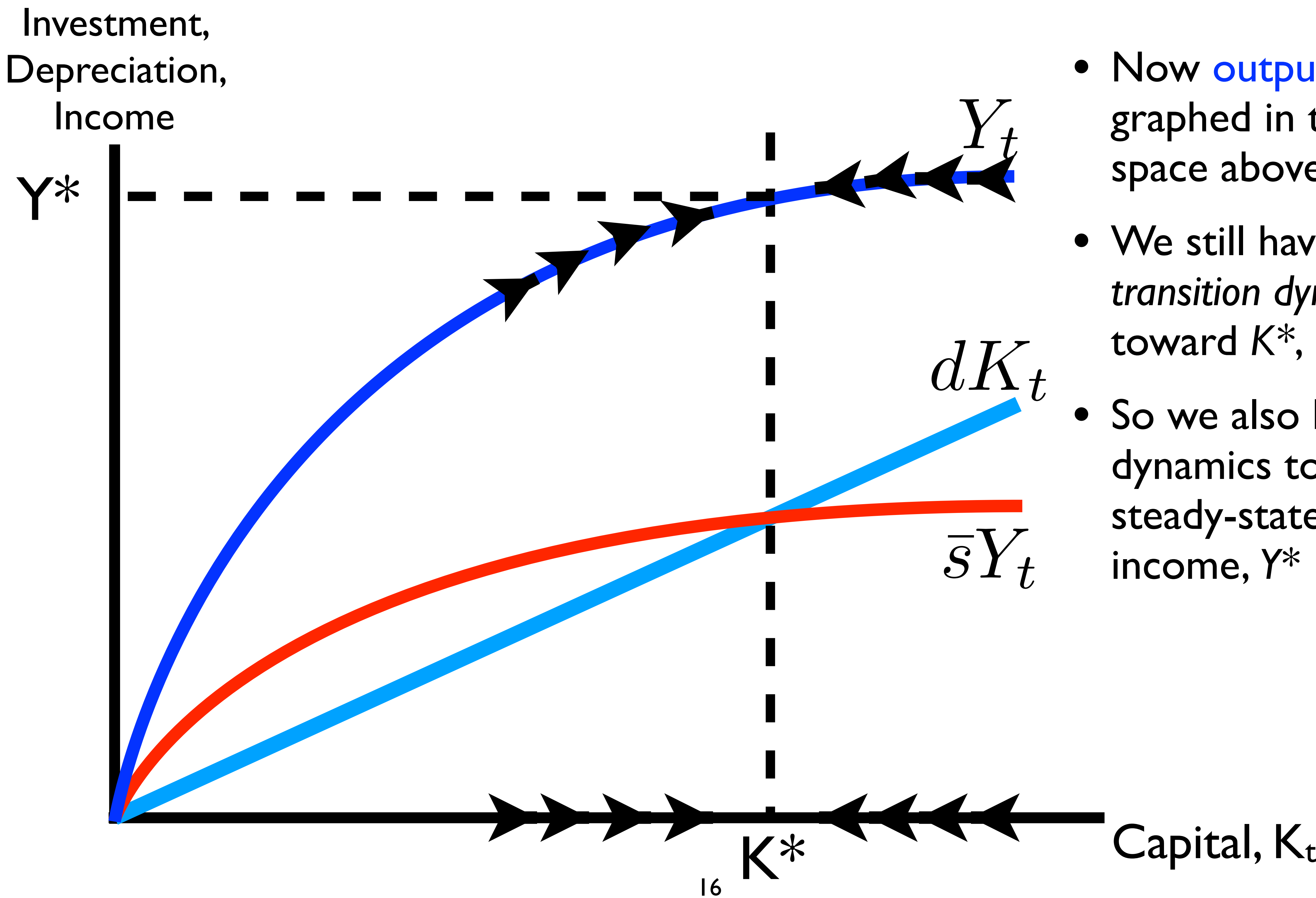
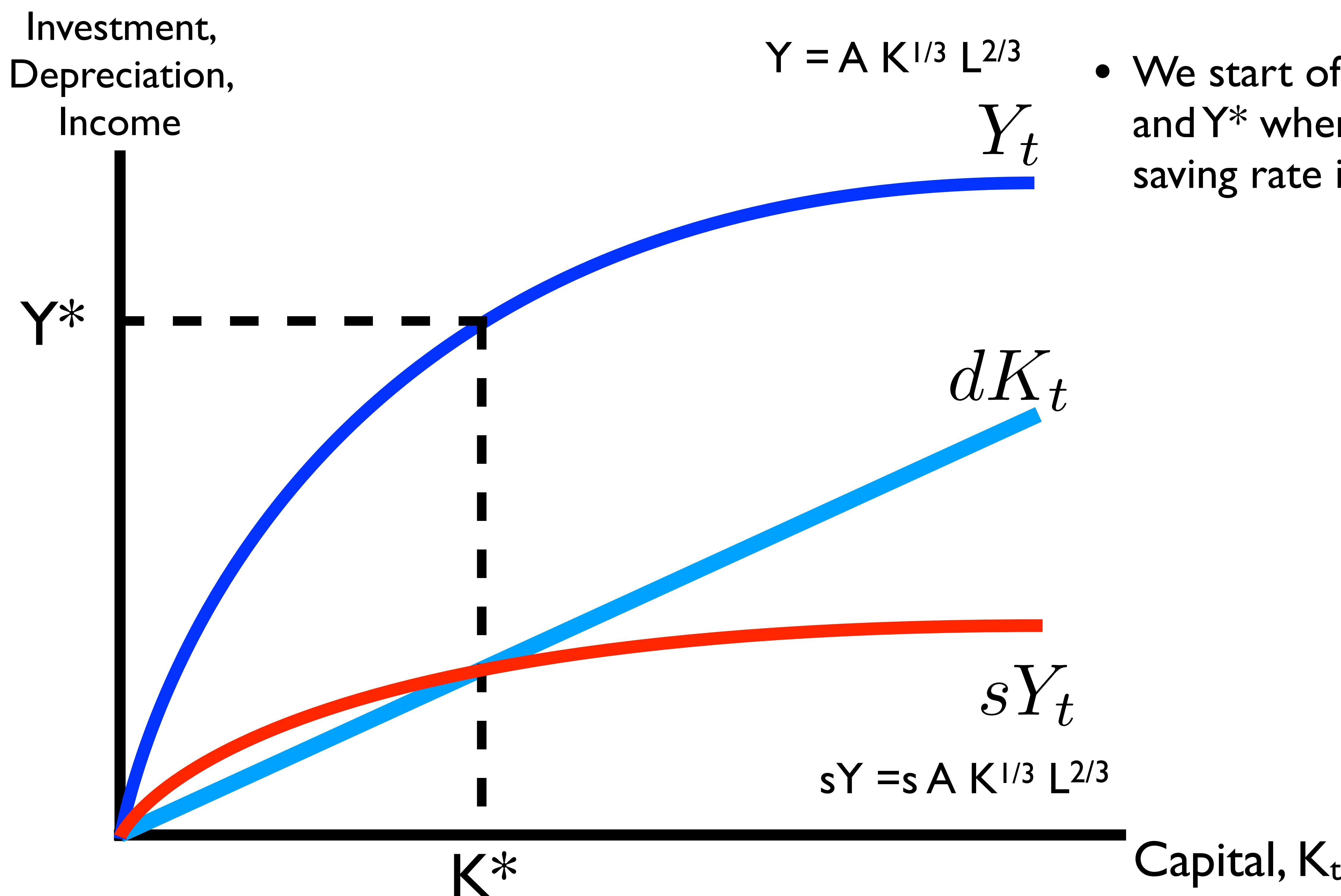


- Now output can be graphed in the space above
- We still have *transition dynamics* toward *K**,
- So we also have dynamics toward a steady-state level of income, *Y**

Suppose the economy begins in steady state at K* but the saving/investment rate rises from s to s' > s



- We start off at K* and Y* when the saving rate is s

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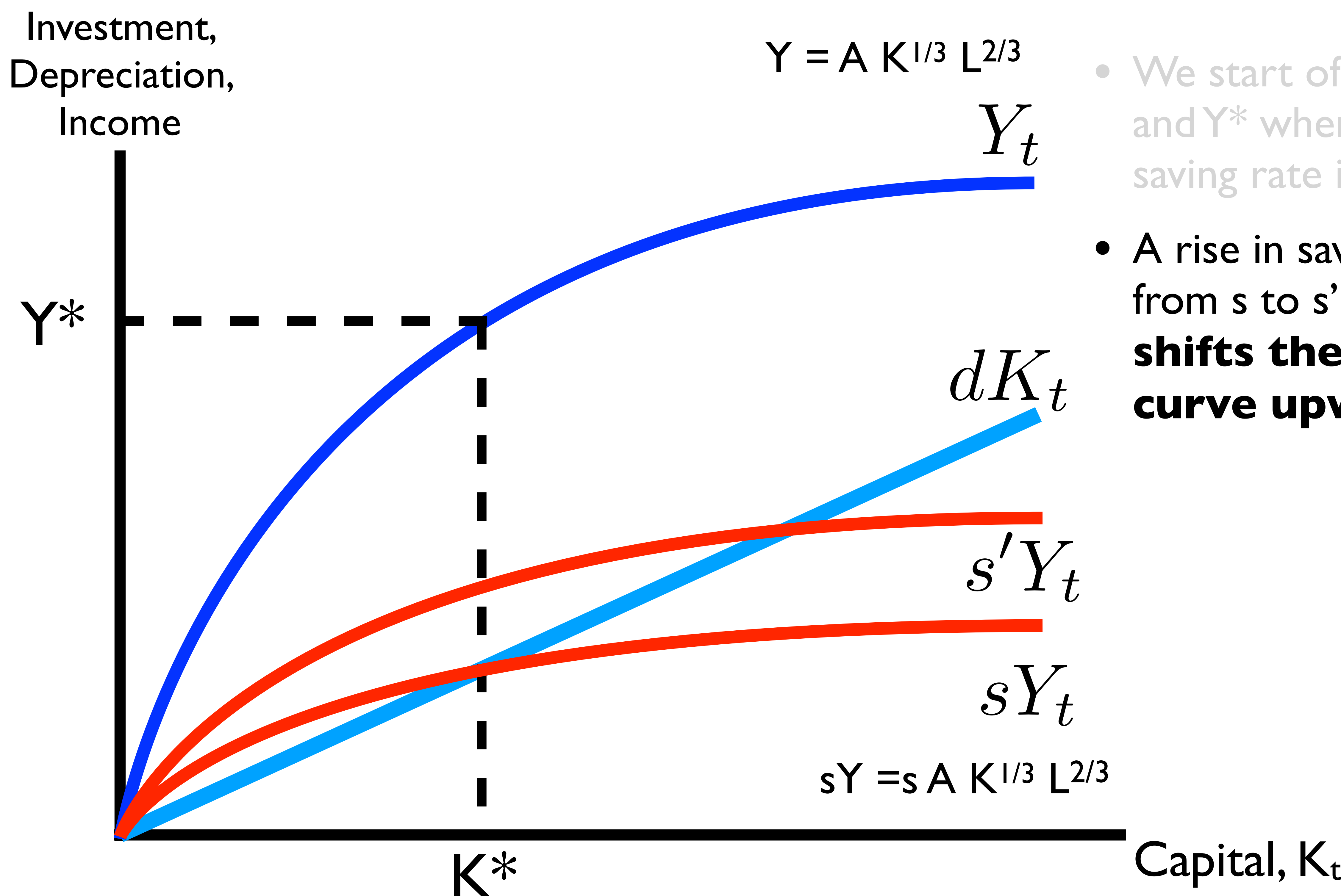

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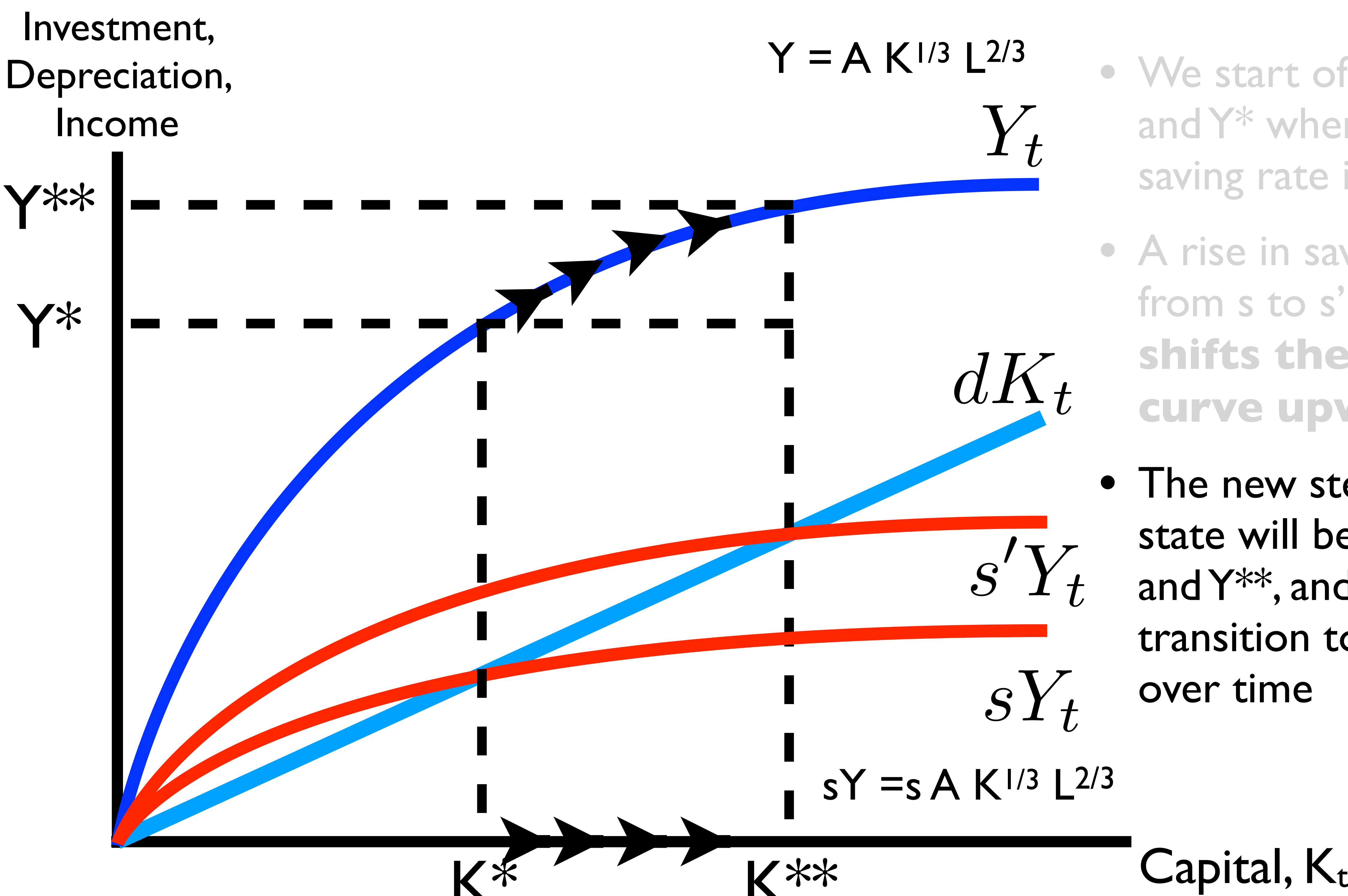
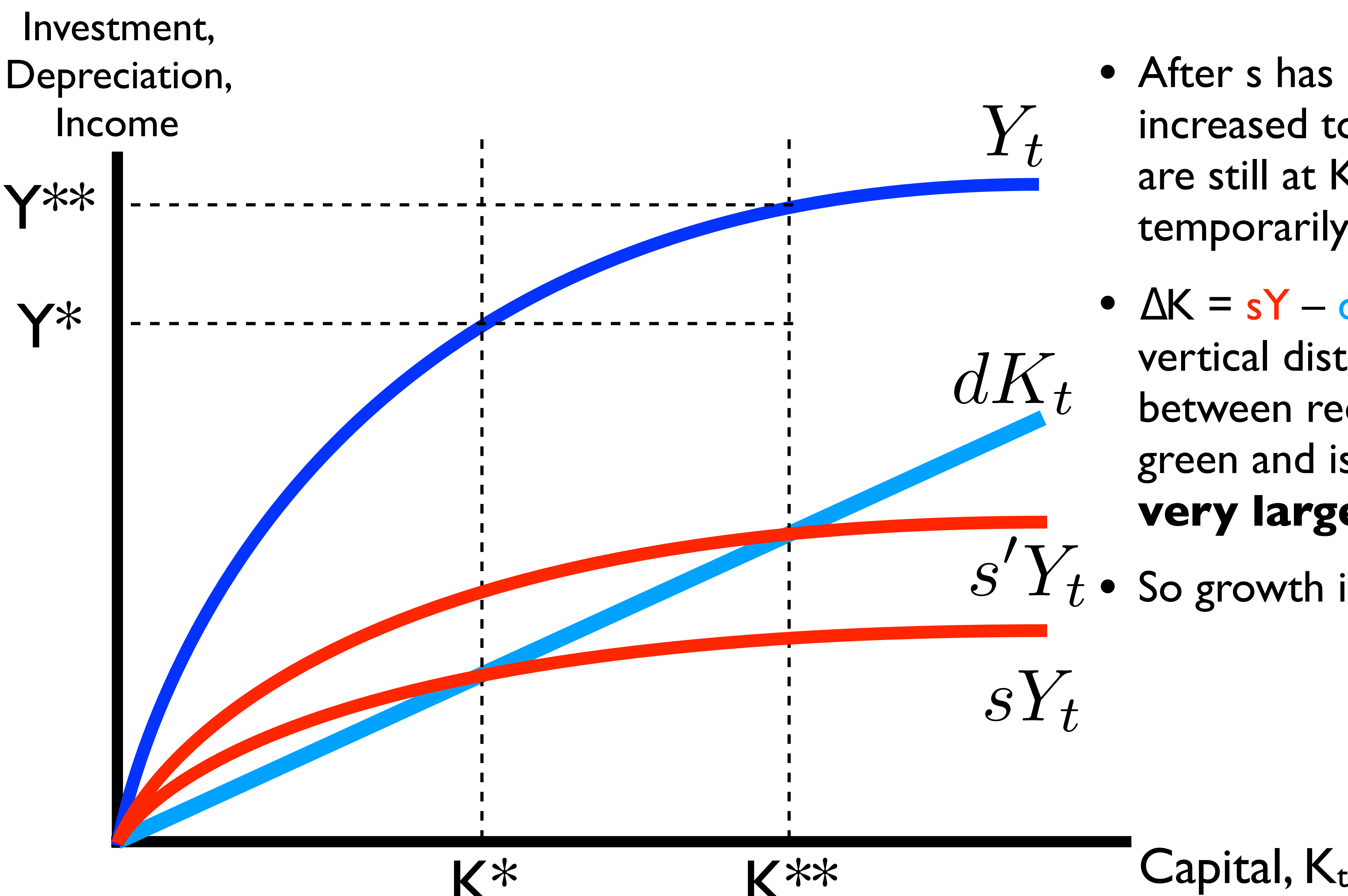


- We start off at K* and Y* when the saving rate is s
- A rise in saving from s to s' > s **shifts the red curve upward**
- The new steady state will be at K** and Y**, and we transition toward it over time

What can we say about the **rate of growth** during the transition to the new steady state, K** and Y**?



- After s has increased to s', we are still at K* temporarily
- ΔK = sY – dK is the vertical distance between red and green and is now **very large**
- So growth is rapid!

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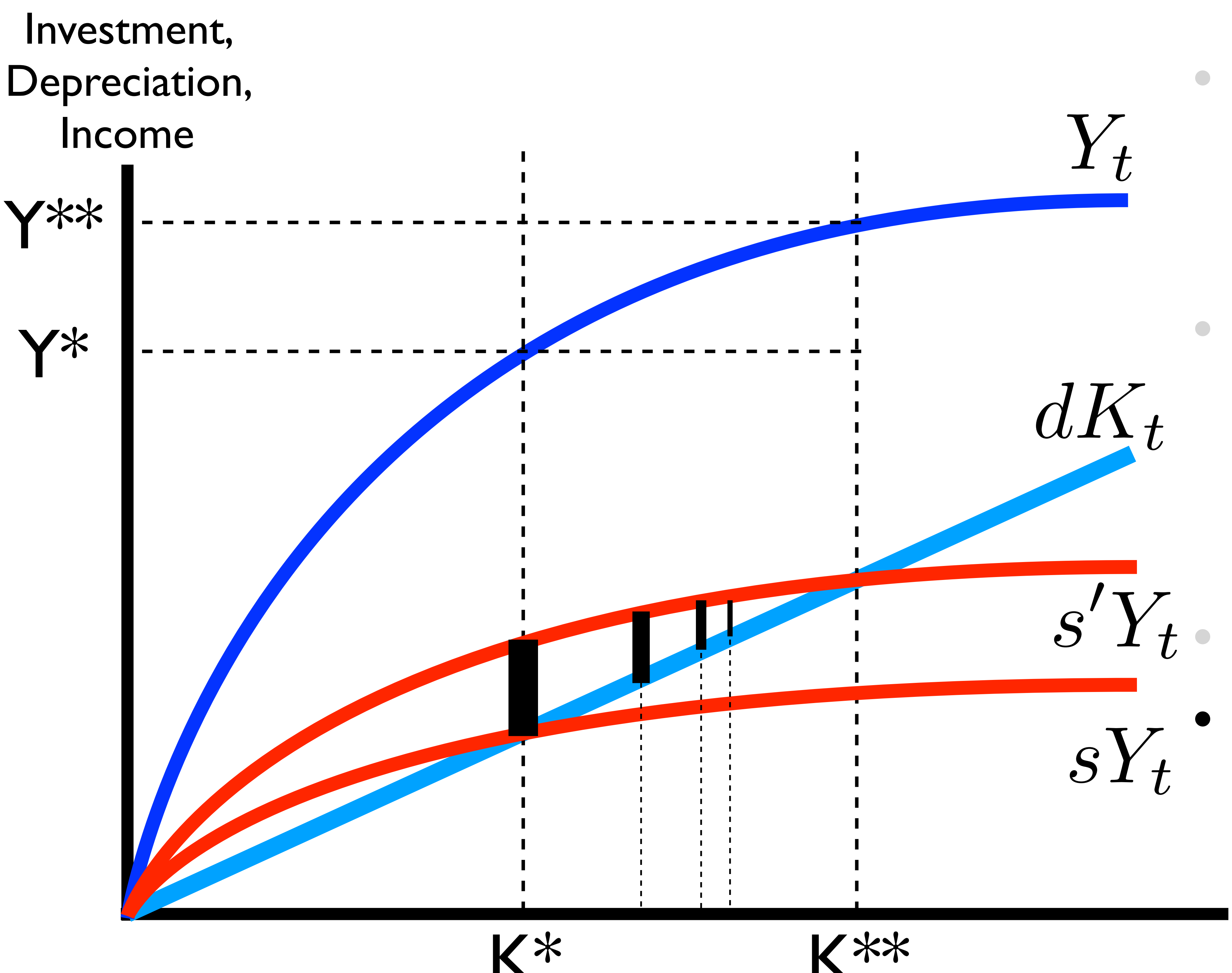
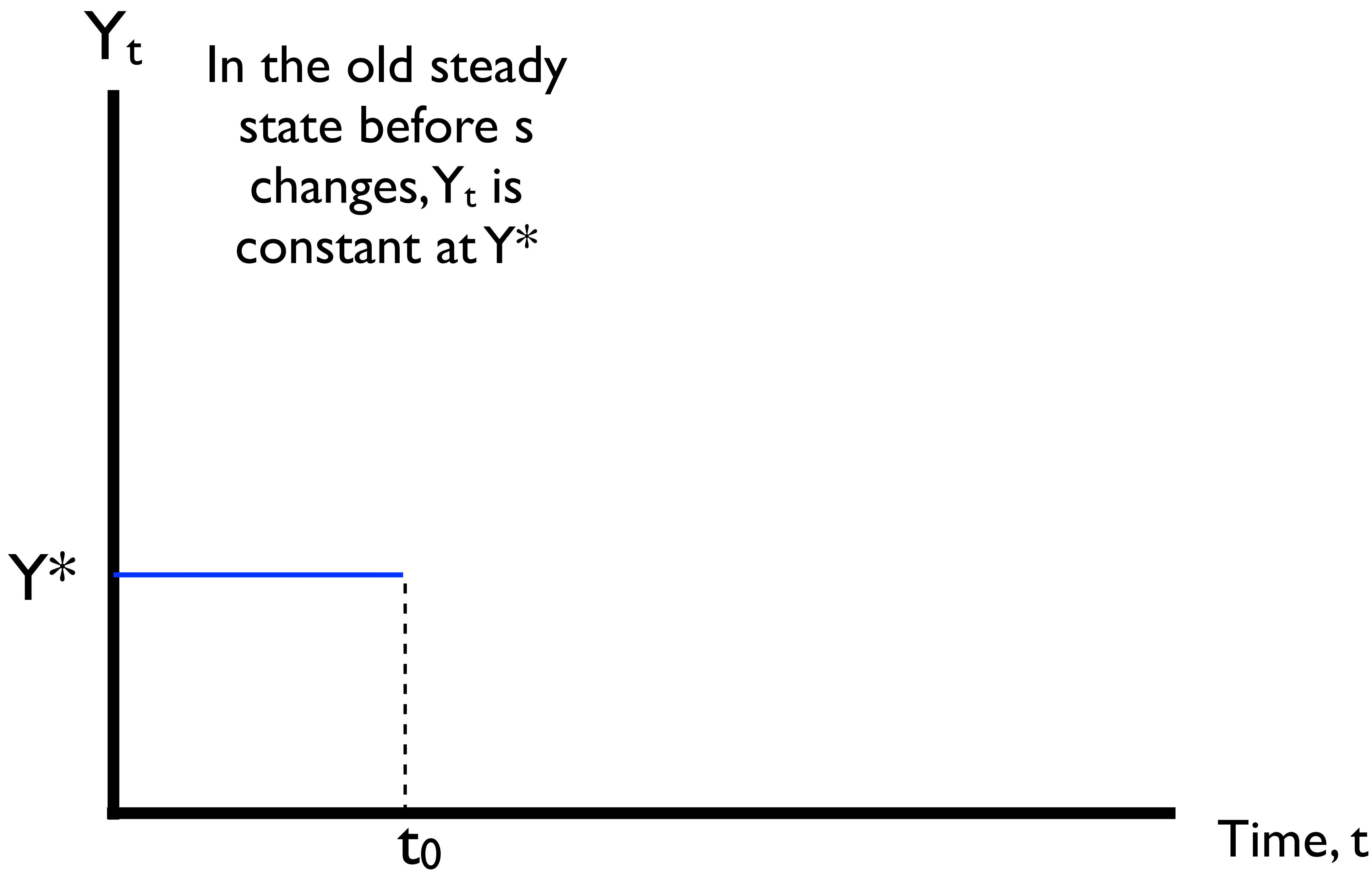


- After s has increased to s', we are still at K* temporarily
- $\Delta K = sY - dK$ is the vertical distance between red and green and is now **very large**
- So growth is rapid!
- As K increases, that distance shrinks, <u>growth slackens</u>

If we calculated and plotted Y_t against time using a ratio/log scale, we would see this:



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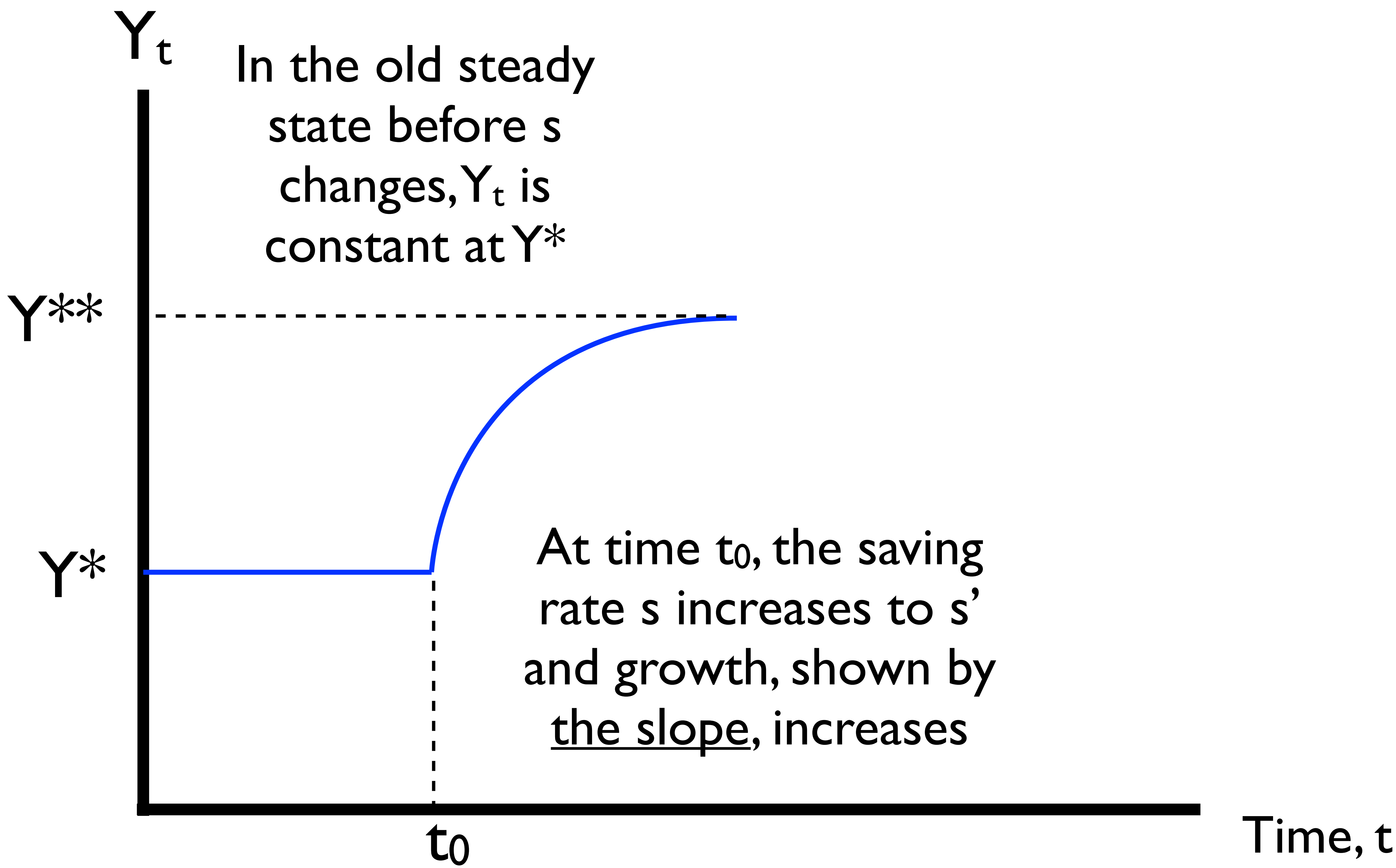

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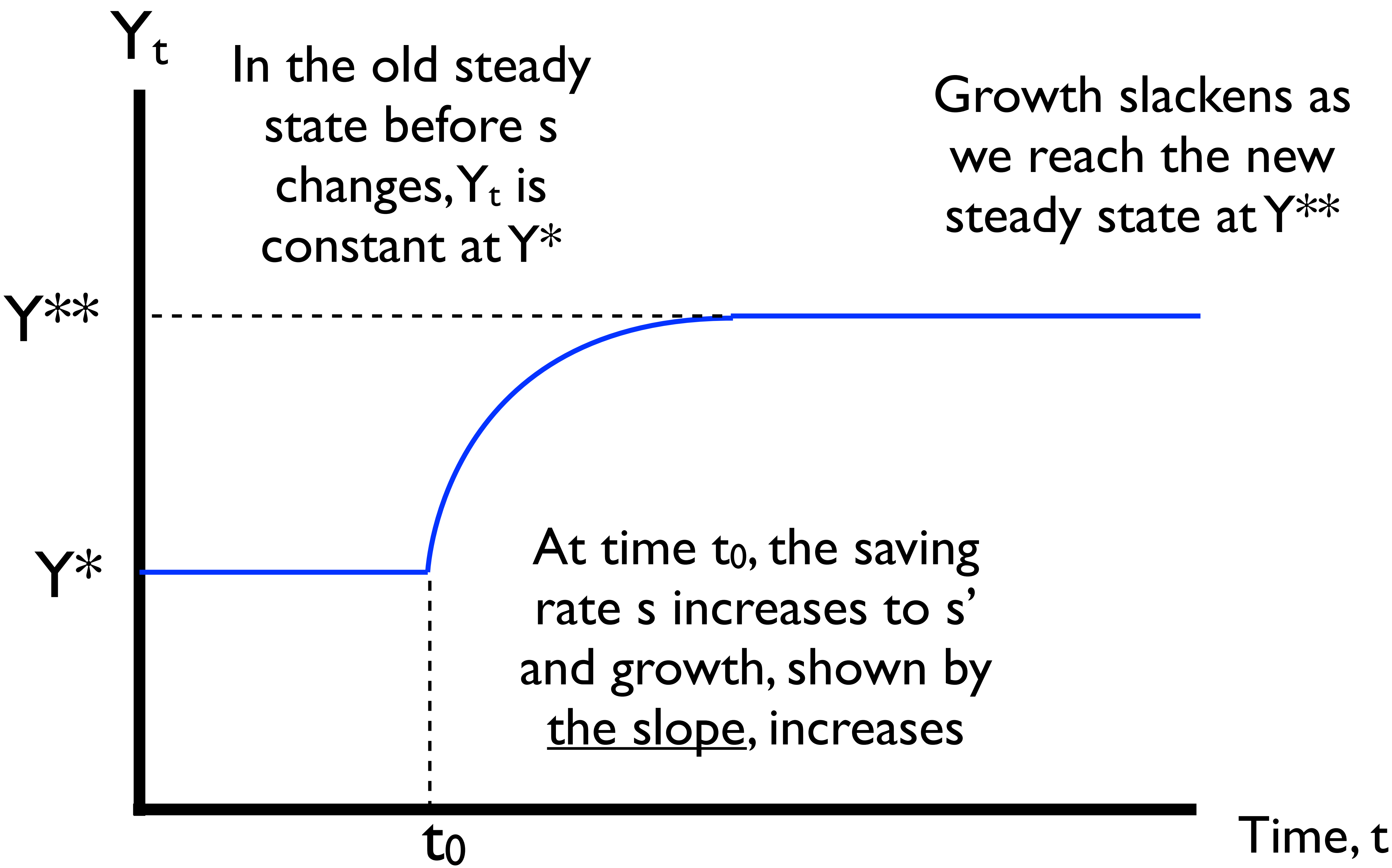
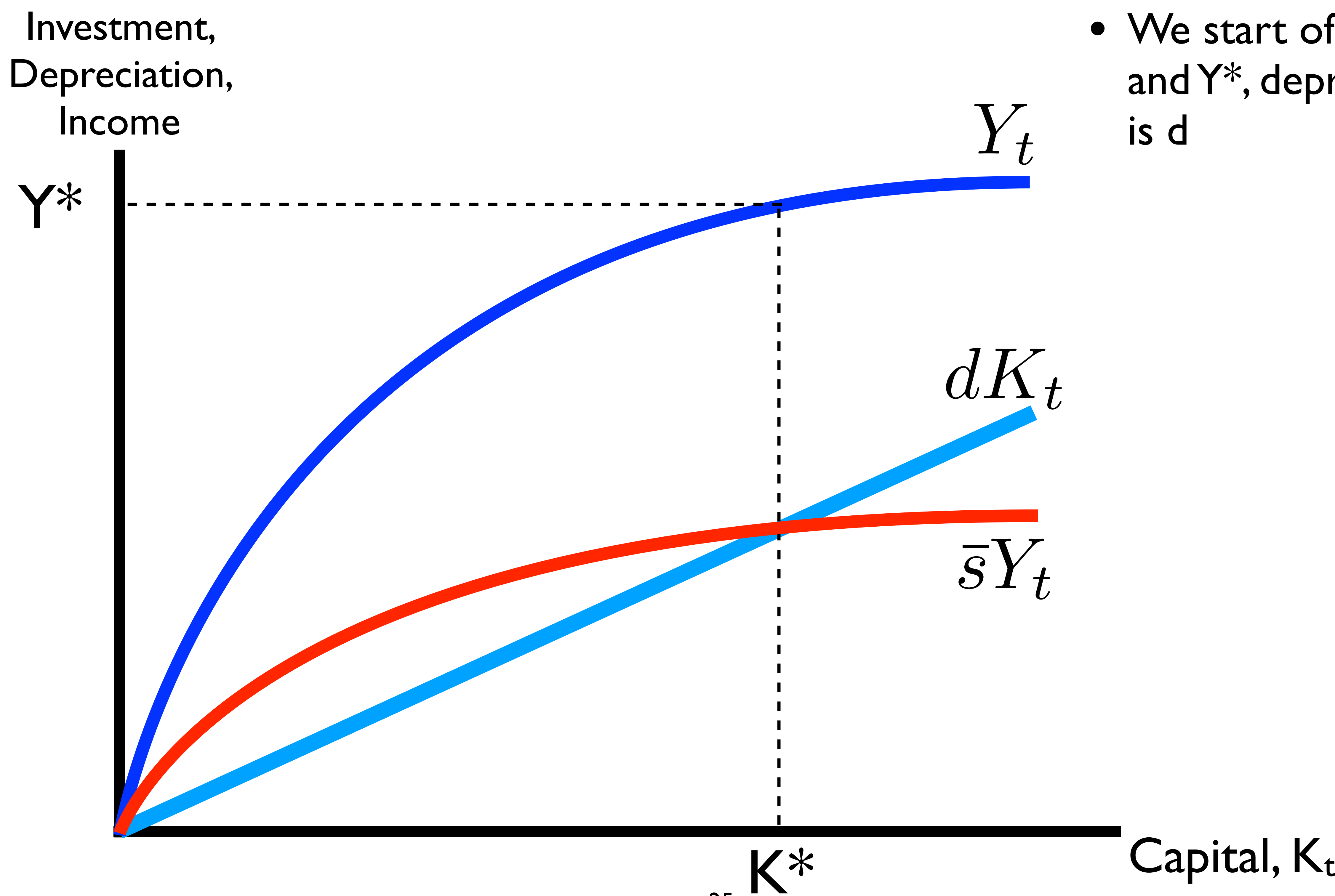


Now let's start over and consider what happens if instead the depreciation rate d rises from d to d' > d



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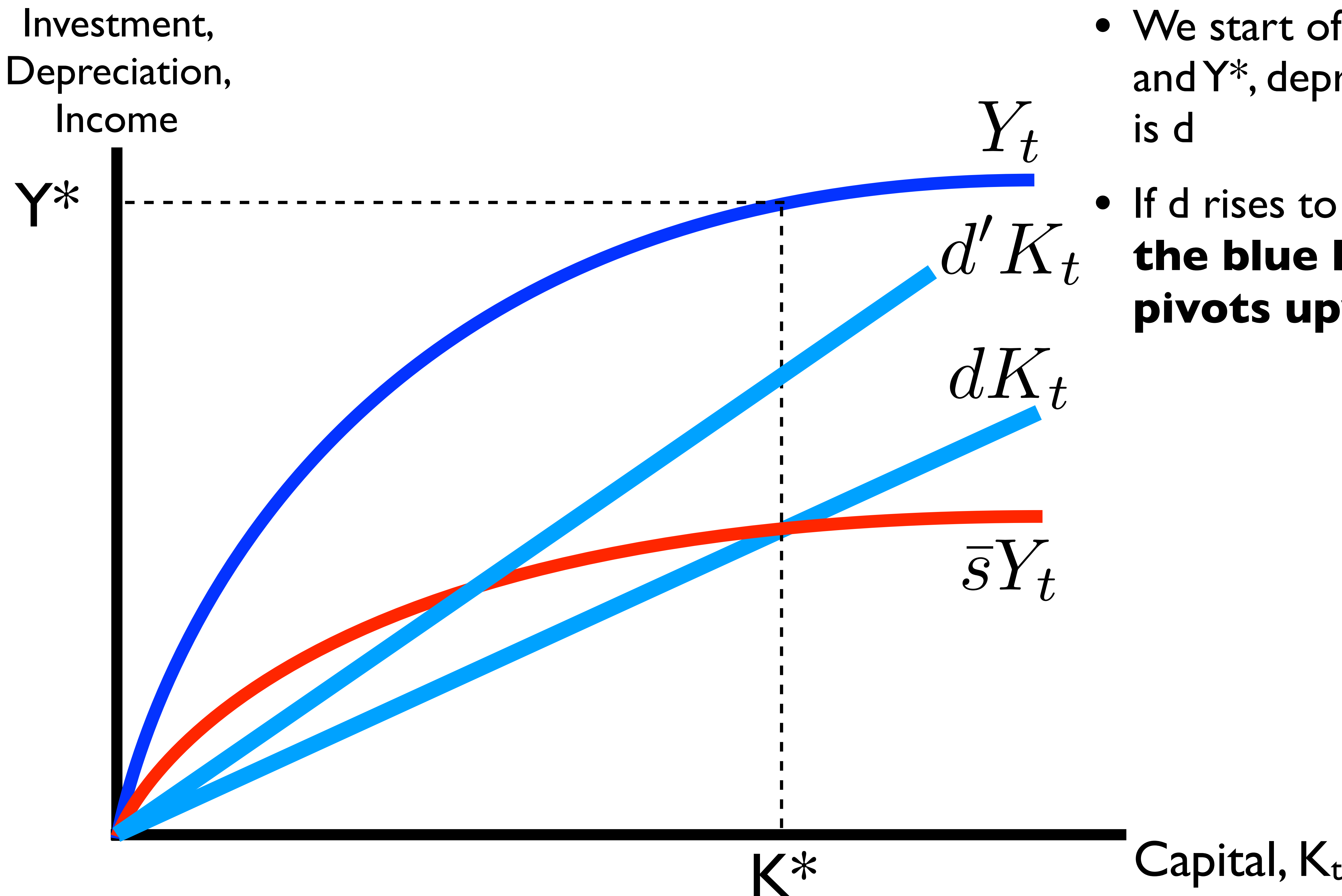

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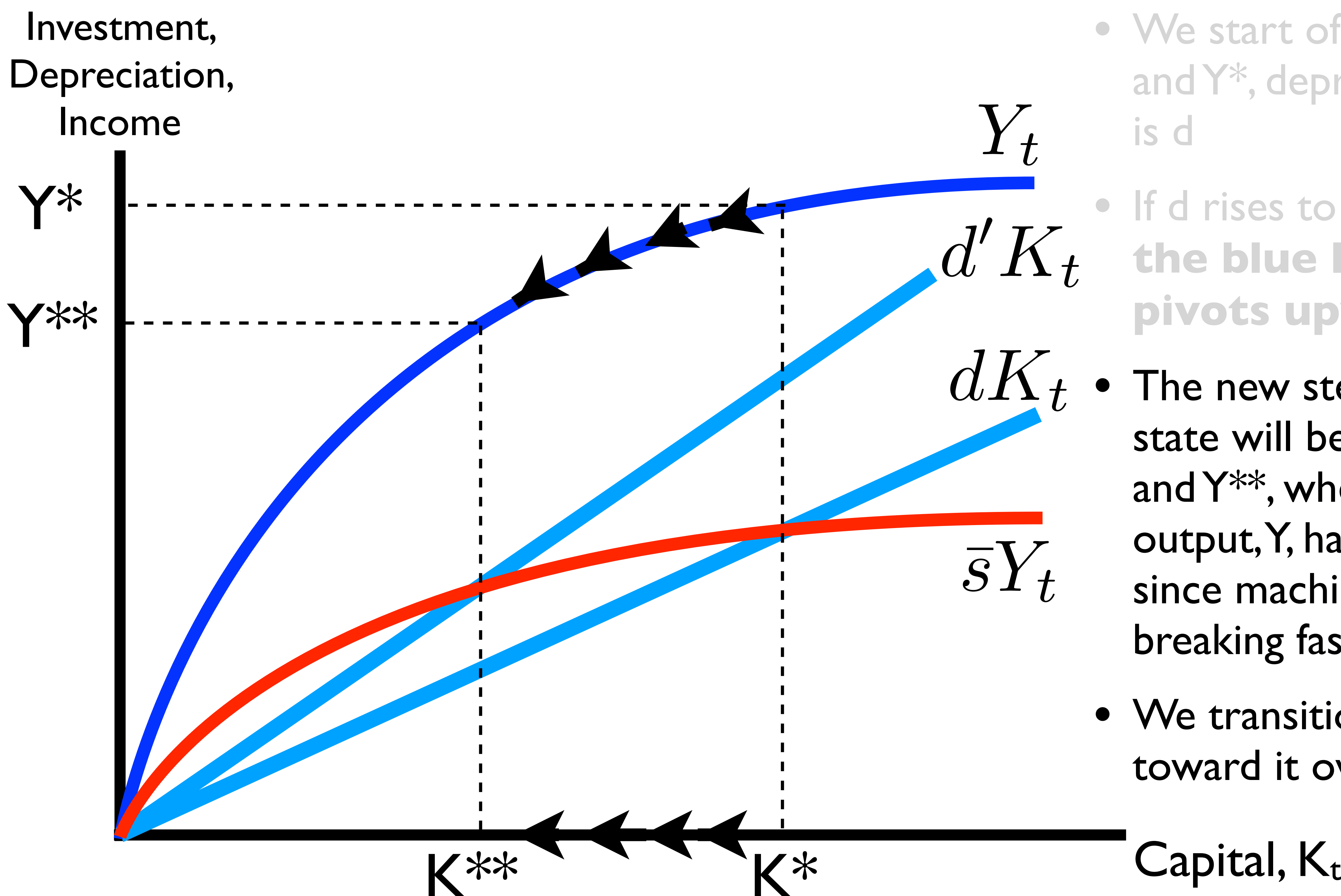
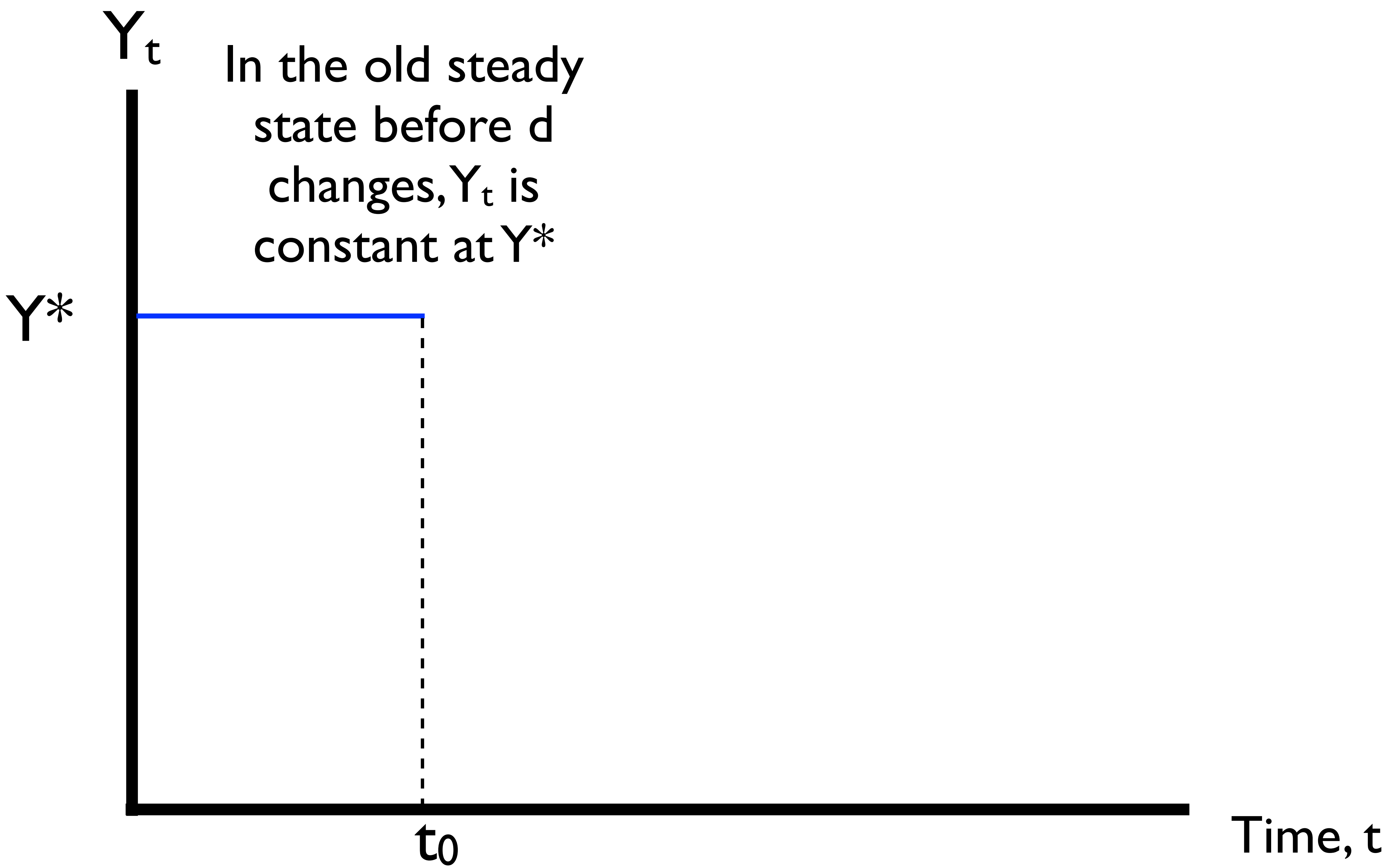


- We start off at K* and Y*, depreciation is d
- If d rises to d' > d, **the blue line pivots upward**
- The new steady state will be at K** and Y**, where output, Y, has *fallen* since machines are breaking faster
- We transition toward it over time

Again, we can see these dynamics by plotting Y_t against time



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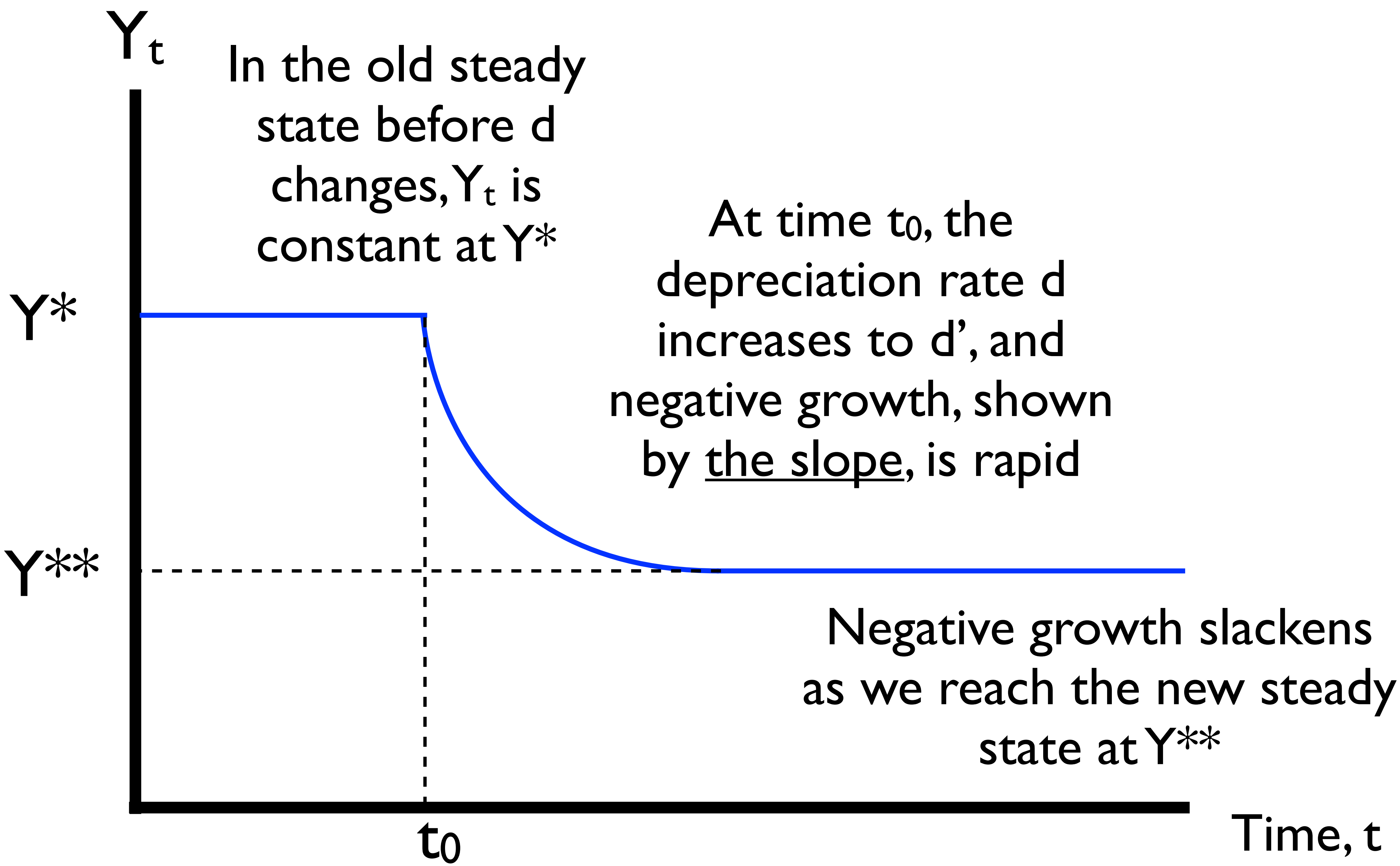


The model explains growth following the <u>destruction of capital</u>

- Why would we be interested in this application?

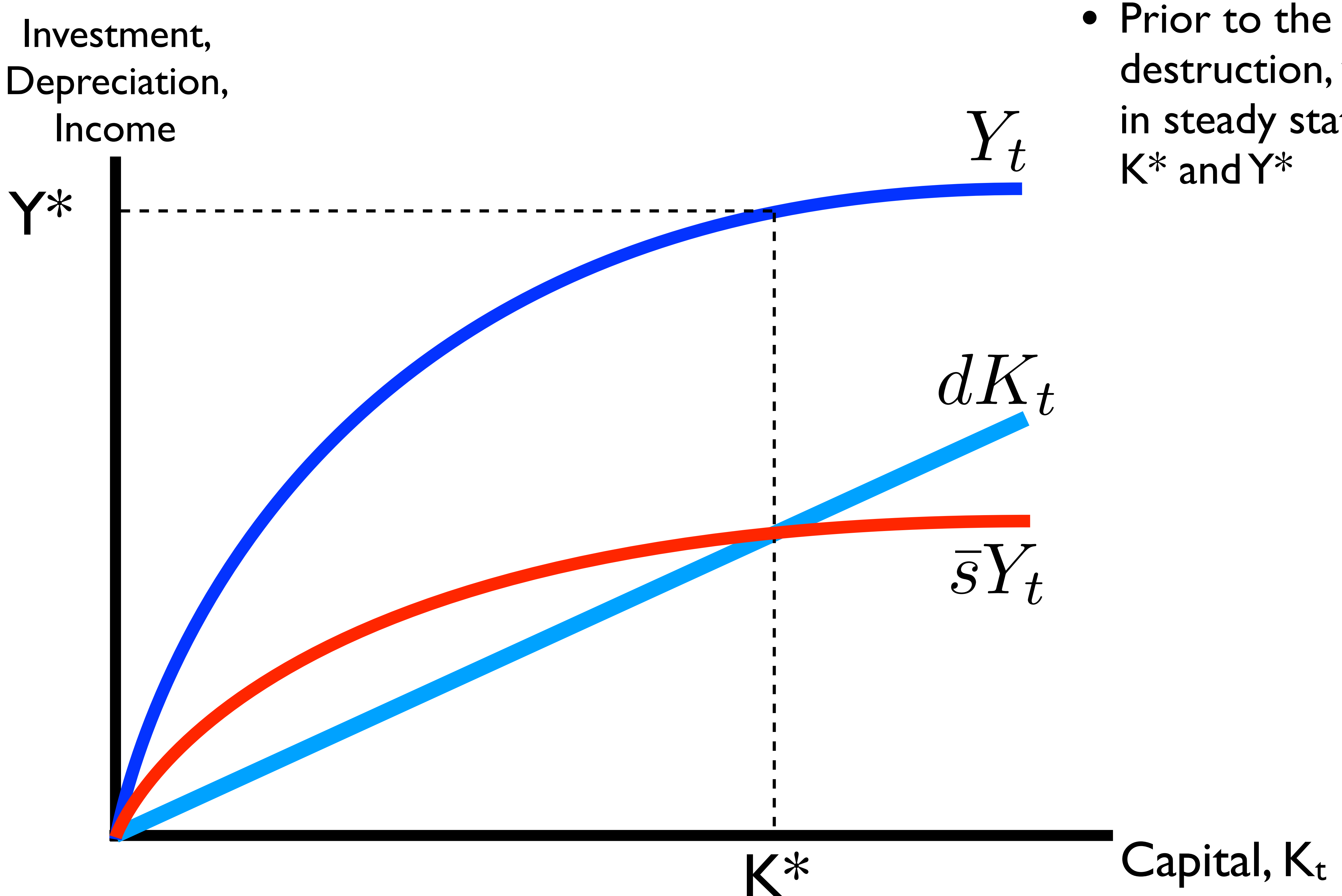
- Important and tragic world events involve capital destruction:

  - **Warfare** destroys productive capital — World War II resulted in the virtual obliteration of German and Japanese industrial capacity

  - **Natural disasters** do too — A noteworthy one in the U.S. was Hurricane Katrina in 2005, which destroyed capital and killed people in New Orleans and along the Gulf Coast

- How do we think about the destruction of capital in the Solow Model? What does it imply about initial effects and recovery?

Destruction of capital per se does not change any fundamentals like s or d; it just removes K



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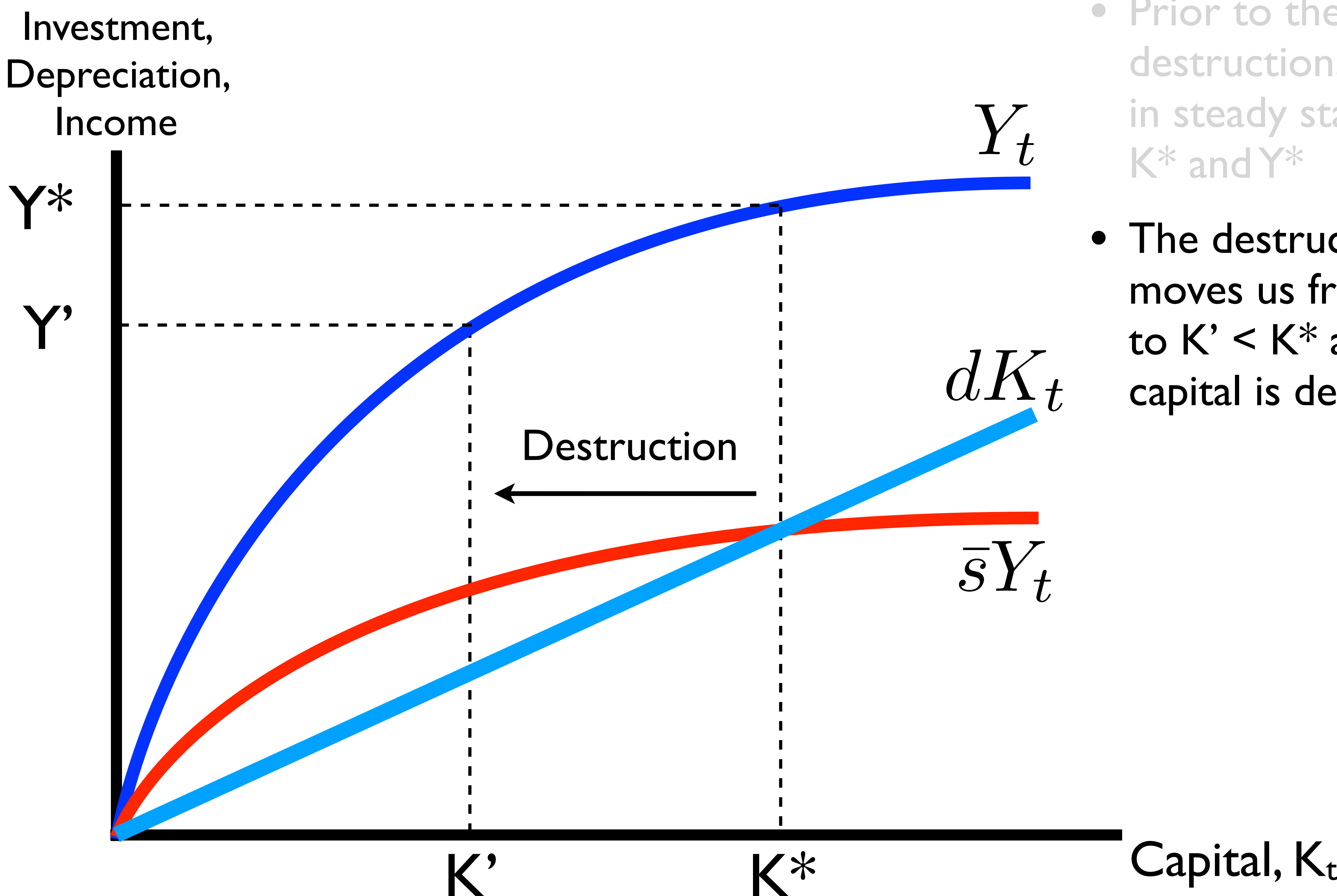

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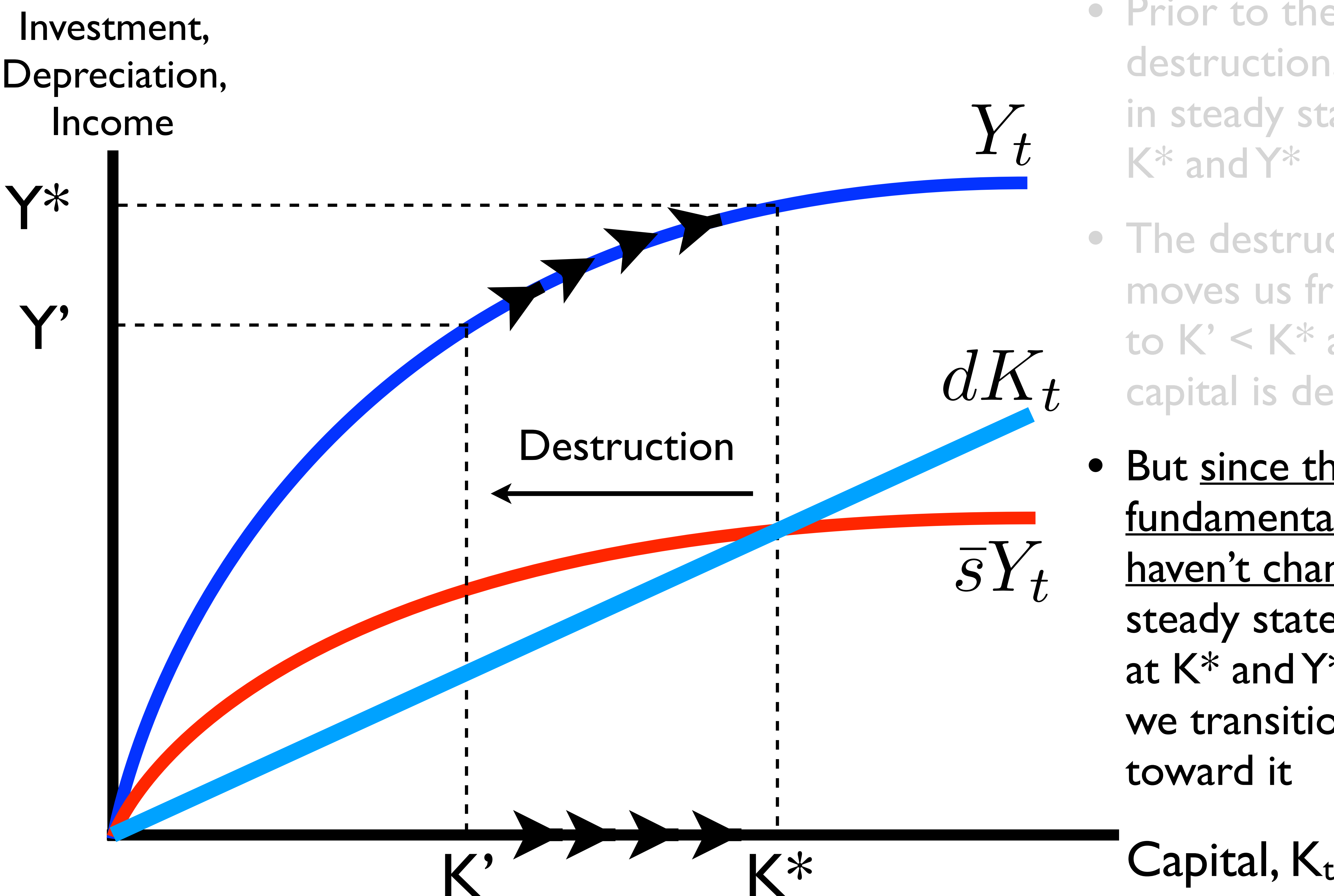
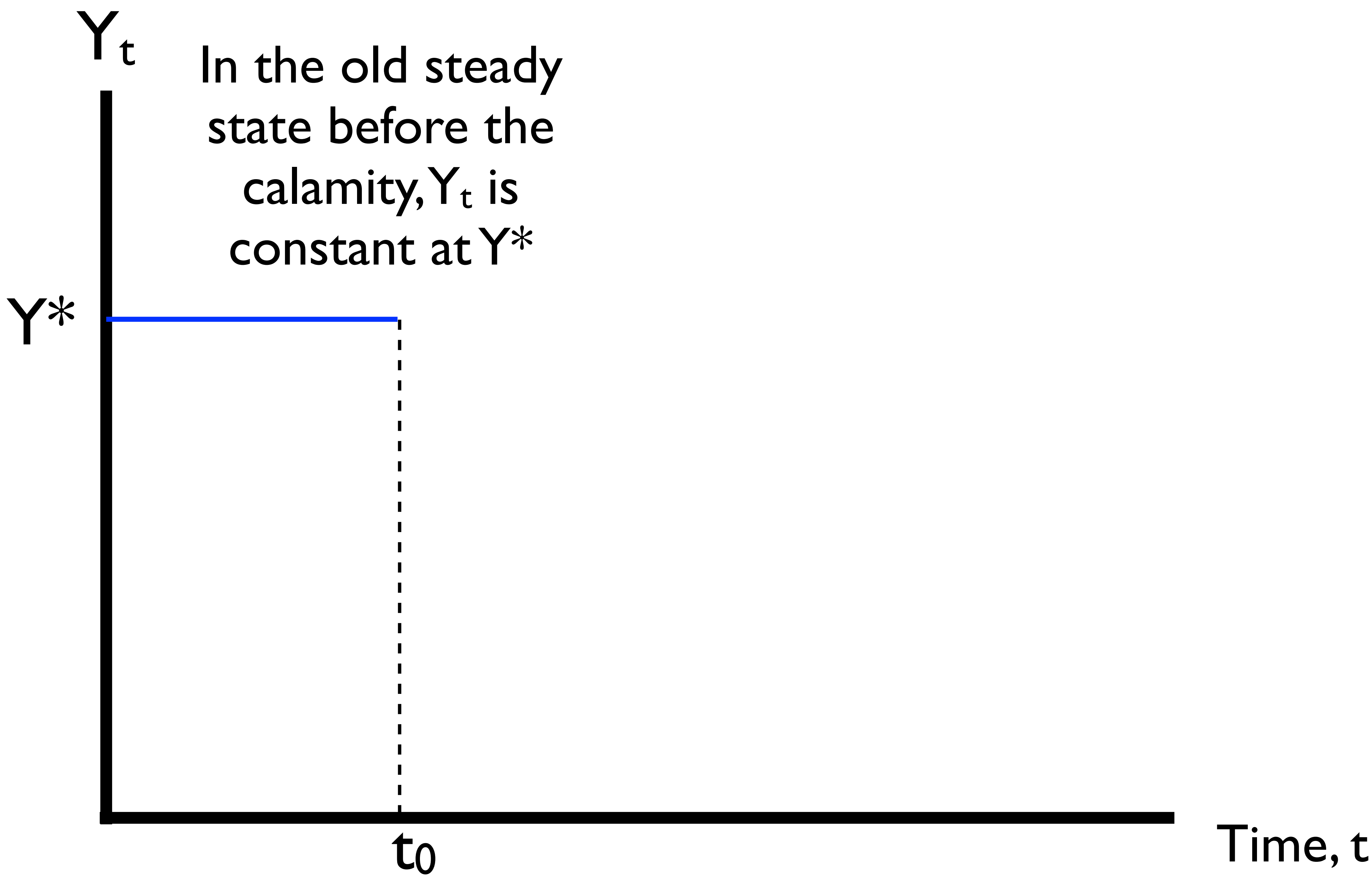


- Prior to the destruction, we are in steady state at K* and Y*
- The destruction moves us from K* to K' < K* as capital is destroyed
- But <u>since the fundamentals haven't changed</u>, the steady state is still at K* and Y*, and we transition back toward it

The graph of Y_t against time shows a *discontinuous jump* in Y_t brought on by the destruction of K_t



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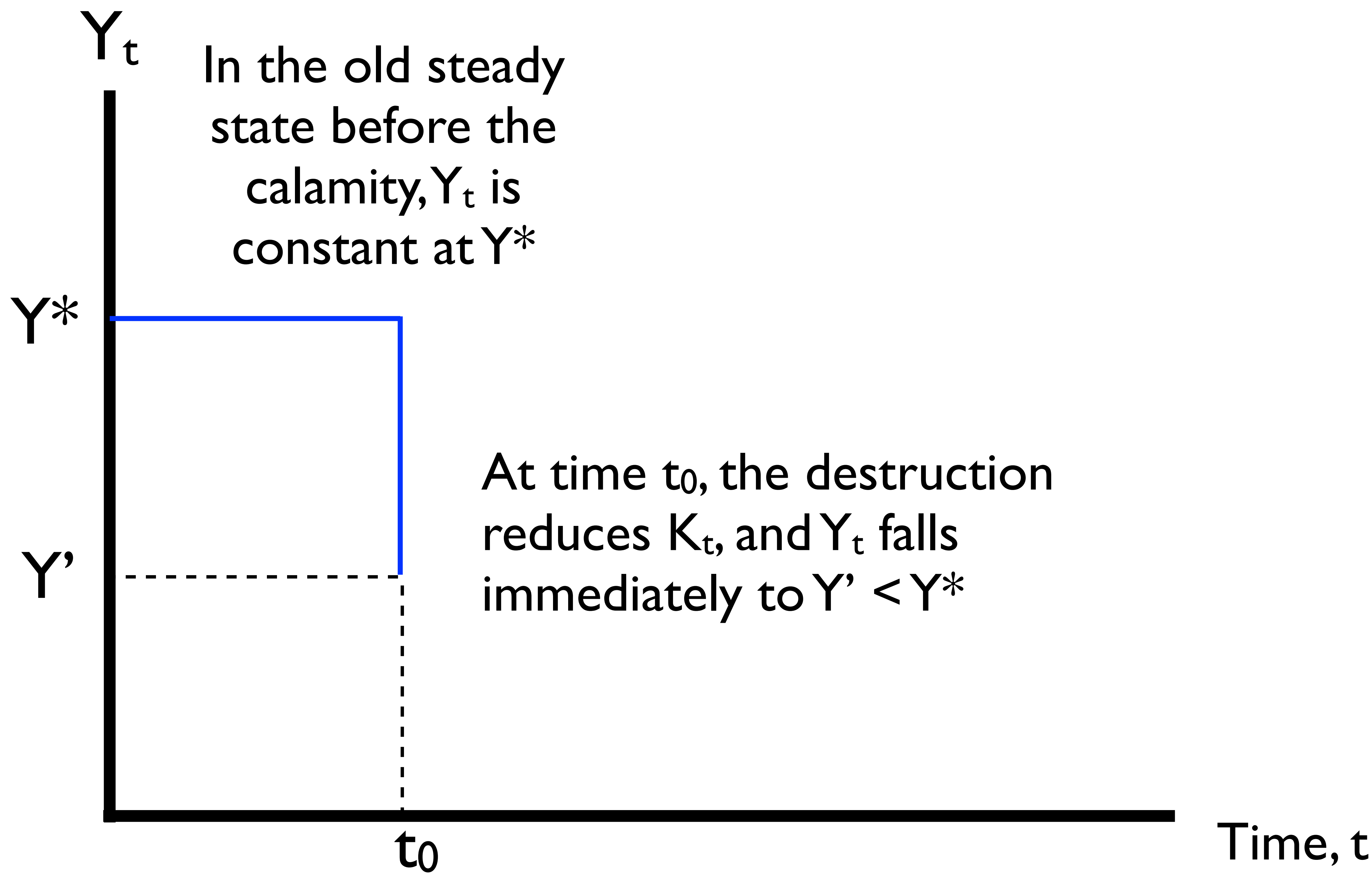

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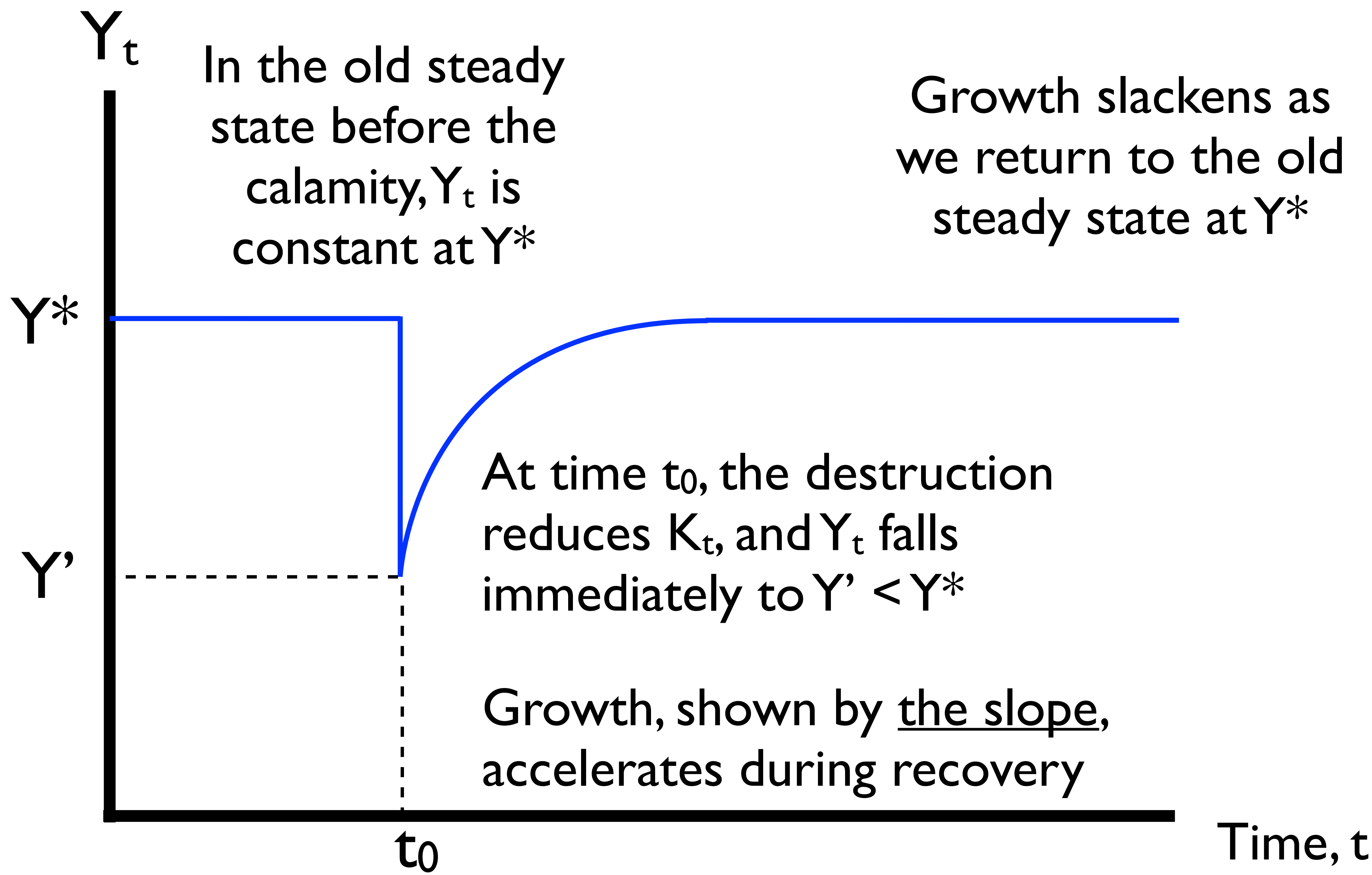


What have we discovered about the relative rate of growth?

- Growth is *faster* when the economy is *further* below its steady-state
- Negative growth, or shrinking, is faster when the economy is further above its steady state — but such cases are rare
- Let's consider the first case: Suppose country A is only 10% below its steady state, while country B is 50% below. Which country will **grow faster?**
- Country B will **grow faster** than country A, even though both are growing, because B is further away from its steady state
- We call this "the principle of transition dynamics"

Why is the principle of transition dynamics useful?

- It describes the postwar experience of industrialized economies pretty well:
  - Japan and Germany grew extremely fast after WWII
  - Reconstruction commenced following the destruction of their capital stocks, much faster than even the U.S.
  - But as these countries approached their steady states, their growth rates fell back to normal
- Today, many developing countries seem to be *trapped* in low steady states rather than growing through transitioning toward new steady states

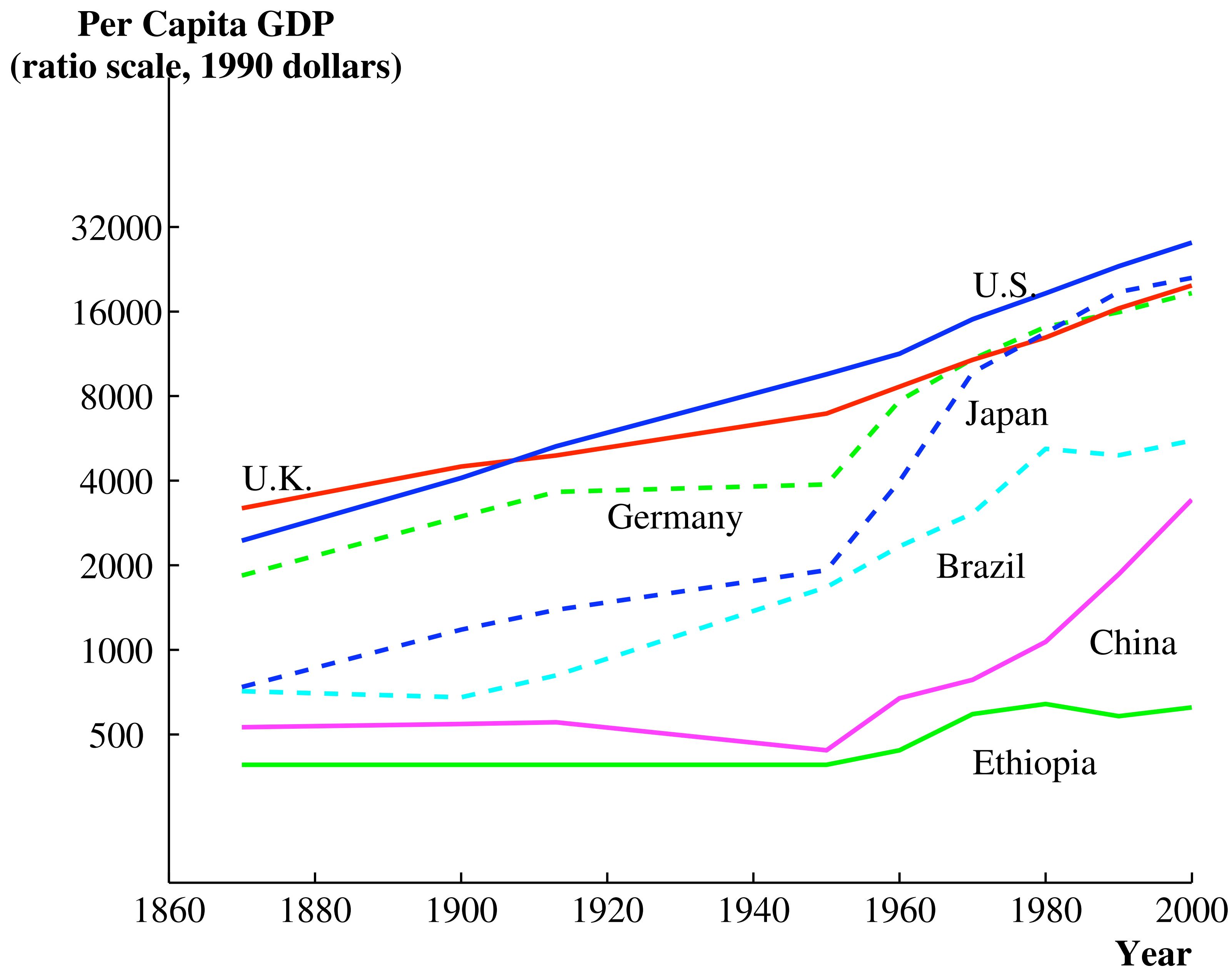
Why are we studying growth?

GDP per capita measures the economic well-being of an average citizen

We want to know why people in some countries are so rich and others elsewhere are so poor!

We also want to know what *sustains* growth in well-being in the long run

So far we have seen that capital (and labor) **can't explain** these very well



Note: Data from Angus Maddison, *The World Economy: Historical Statistics* (Paris: OECD Development Center, 2003). Observations are presented every decade after 1950 and less frequently before that as a way of smoothing the series.

Why can't **capital** explain growth over time or differences between countries?

- There are *diminishing* returns to capital
  - The more machinery you acquire, the less the last machine really helps you. Why?
  - Capital goods are ***rival*** — if you use a machine in one place, you *can't* use it elsewhere
- What does not encounter diminishing returns?
- **IDEAS** and techniques, which can be spread broadly

What has the Solow Model revealed?

- Through saving and investing balanced against depreciation, the economy moves to a **steady state**, regardless where it began

- We reach a steady state because:

  - There are diminishing returns to capital: less Y_t per additional K_t

  - That means new investment is also diminishing: less $sY_t = I_t$

  - But depreciation is NOT diminishing; it's a constant share of K_t

- After we reach the steady state, there is **no** long-run growth in Y_t (unless L_t or A increases): Saving and investing, a.k.a. capital accumulation, cannot produce long-run growth!

If productivity is still so important (and so far, unexplained), what good is the Solow model?

- It explains real-world capital-output ratios (K_t/Y_t) well, so parts of it match reality. With it we understand **capital accumulation**

- It provides new insights into growth:

  - Changes in the "parameters" — the exogenous variables in the model like the saving rate, s, and the depreciation rate, d — change the steady state and induce economic growth in the transition to the new steady state

  - This reveals *the principle of transition dynamics*: during the transition to the steady state, growth will be faster for the country that starts further from its steady state